

Borrowing From Government Owned Banks & Firm's Liquidation Risk *

Ankit Kariya

Indian Institute of Management, Bangalore [†]

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Abstract

Government Owned Banks (GOBs) have other explicit or implicit objectives apart from profit maximization. In this paper, I study whether this affects the liquidation risk of firms borrowing from GOBs. Using the natural experiment of securitization reform in India that increased firms' liquidation risk, I find that the firms borrowing exclusively from GOBs did less reduction in secured debt usage compared to other firms. In the cross-section, the effect is more substantial in the subsample of firms that are more likely to face financial distress. These results suggest that borrowing from GOBs have less liquidation risk.

Keywords: Government Owned Banks, Secured Debt, Securitization Reform, Liquidation Risk, Distress, India

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[†]Email: ankitkumark16@iimb.ac.in; Address: IIM Bangalore, Bannerghatta Road, Bangalore, India

1. Introduction

Liquidation risk arises due to the non-linear nature of debt and equity claims. The creditors, because of their concave payoff, have a bias towards liquidation, i.e., they might liquidate firms that have higher continuation value. While the equity holders have continuation bias, arising from the convex nature of their financial claims and the non-contractible private benefits they enjoy. In a creditor friendly bankruptcy system, creditors decide whether to continue or liquidate a firm facing financial difficulties. That increases the liquidation risk for the firms that have high liquidation value. Anticipating this, firms use lower debt in creditor friendly countries (Acharya and Subramanian (2009); Acharya, Sundaram, and John (2011b); Vig (2013)).

In emerging markets, government owned banks (henceforth GOBs) have a large share in credit markets. A significant number of firms borrow from them. From exiting literature, we do not know whether borrowing from GOBs has any effect on the firm's liquidation risk. I explore this question in the paper using a natural experiment of securitization reform in India that increased liquidation risk for the firms using secured debt financing (Vig (2013)). Using difference-in-difference methodology, I document that in response to this shock, firms borrowing exclusively from GOBs (henceforth GOB firms) reduce their usage of total debt and secured debt relatively less.

In the theoretical models (Acharya et al. (2011b); Vig (2013)), the profit maximization objective of the lender drives the liquidation bias. But the government, through government owned enterprise, does not participate in the market solely to make profits.

There are three theories of government ownership of enterprises. First, the social view suggests that government owned enterprises, like any other intervention in the markets, are created to address the market failure whenever the social benefits of government ownership are more than the costs (Stiglitz (1993)). According to this view, the GOBs facilitate economic development and improve the general welfare. Second is the agency view. It agrees with the social view that government owned enterprises are created to maximize the social

welfare, but may give rise to corruption and misallocation of funds (Banerjee (1997)). They maximize multiple non-measurable objectives. That leads to agency problems and weak managerial incentives. Lastly, the political view postulates that government owned enterprises are a medium for pursuing politicians' goals such as funding the favored enterprises and maximizing the employment (Shleifer (1998)).

The empirical literature on bank efficiency suggests that GOBs are comparatively less efficient (Micco, Panizza, and Yanez (2007)). They have a relatively lower return on assets and interest margin and higher overhead costs. The inefficiency of GOBs in itself cannot distinguish between the theories of state ownership of enterprises: it is not obvious whether GOBs are less efficient because they maximize broader social objectives, or because of internal agency problems, or because they serve political interests (Sapienza (2004)). There is some evidence that GOBs' lending is less cyclic (Coleman and Feler (2015); Bertay, Demirg-Kunt, and Huizinga (2015); Micco and Panizza (2006)), and so they play some role in reducing the severity of recessions which is consistent with the social view. But several single country studies examining detailed bank lending behaviour while dealing with politically connected firms or during elections supports the political view (Sapienza (2004); Carvalho (2014); Ding (2005); Khwaja and Mian (2005); Cole (2009); Kumar (2019); Cull, Li, Sun, and Xu (2015)).

However, all three theories agree that the GOBs are unlikely to have only a profit maximization objective. They might have a social objective such as facilitating economic growth and maximizing employment by providing credit to financially constrained firms, or they could be serving the personal goals of their managers or politicians. Given these other objectives, it is intuitive that GOBs are relatively less likely to push for liquidation in response to borrowers' financial difficulties, even if it's optimal from a profit maximization point of view. For example, under the social view, GOBs care about social welfare so they will not force inefficient liquidations. Under the agency view, the manager of a GOB might not liquidate the firm even though it's optimal for the bank because of the manager's other benefits such as future job prospects in the private sector. Concerning the political view, GOBs

would not liquidate firms for political reasons such as employment preservation or firms are politically connected. Anecdotal evidence does suggest that GOBs are less stringent in debt recovery compared to private banks. For example, former governor of Reserve Bank of India, Raghuram Rajan notes in his speech titled “Resolving Stress in the Banking System” in June, 2016¹:

“Moreover, as a project went into distress, private banks were sometimes more agile in securing their positions with additional collateral from the promoter, or getting repaid, even while public sector banks continued supporting projects with fresh loans. Promoters astutely stopped infusing equity, and sometimes even stopped putting in effort, knowing the project was unlikely to repay given the debt overhang”

The above discussion and a common empirical observation that GOBs have a higher level of non performing assets (NPAs) than private banks suggest that firms borrowing exclusively from GOBs will have lower liquidation risk. But this is not obvious. Comparatively lower NPAs of private banks could be due to cherry picking of good quality borrowers by them. There is some evidence of foreign banks doing that in India (Gormley (2010); Berger, Klappper, Peria, and Zaidi (2008)). Further, “other objectives” of GOBs need not translate into less liquidation bias. It could just mean that they help the government in various developmental and administrative activities². And they lend to a different set of borrowers. The social view might translate to lending to credit rationed small industries, while the agency and the political view might mean giving priority to “friends”.

Furthermore, GOBs in India have one more incentive to make strict recoveries. In India, GOB employees are considered as a public servant and, so they can be investigated for cor-

¹Reserve Bank of India website: m.rbi.org.in

²Indian government owned banks do serve as administrative agents of the government for various developmental and administrative activities. In a recent large scale financial inclusion program launched by India’s government for giving bank accounts to unbanked citizens, GOBs opened a disproportionately high share of new accounts (Chopra, Prabhala, and Tantri (2017)).

ruption. In the past, some GOB officers have been investigated for corruption in lending³. Commentators argue that this has created fear in bankers and affected genuine lending and debt restructuring decisions⁴. This fear of investigation could increase the GOB manager’s likelihood of choosing liquidation rather than reorganization in the event the borrower’s distress. That could increase liquidation risk for GOB firms. Given these competing arguments, the effect of borrowing from GOBs on firms’ liquidation risk is not apparent, and therefore, it is an empirical question.

Since the firm’s debt usage is decreasing function of liquidation risk, an obvious way to answer this question would be to compare the debt of firms borrowing from GOBs to that of firms borrowing from other banks. However, the difference in debt level depends on multiple factors, and due to the non-random assignment of firms to banks, the comparison will be biased. Therefore, we need exogenous variation in liquidation risk to study the behavior of comparable firms borrowing from GOBs and other banks. This paper uses securitization reform in India as a source of exogenous variation in liquidation risk. India provides an ideal set up for studying this question due to two reasons. First, the Indian economy has a substantial involvement of GOBs in the banking industry. In 2016-17, the share of GOBs in total credit was 68.5%⁵. The government appoints the board and CEO of GOBs. They periodically report to the finance minister and take instructions on the priorities of the government. Second, India has done important changes in its legal structure that has improved creditor rights. In this paper, I exploit one of these changes, the enactment of the Securitization and Reconstruction of Financial Assets and Enforcement of Security Interests Act of 2002 (henceforth SARFAESI Act).

SARFAESI Act substantially increased the powers of secured lenders. It allowed them to speed up the process of seizing and liquidating the collateral of defaulting borrowers.

³“Banks plan to back staff against inquiry agencies”, Economic Times, 6th August, 2018

⁴Rebello, Joel. “No penalty for not taking decisions, but sword for taking one: why files don’t move in banks” Economic Times, July 18, 2019. Accessed November 11, 2019. <https://prime.economictimes.indiatimes.com/news/70269488/fintech-and-bfsi/no-penalty-for-not-taking-decision-but-sword-for-taking-one-why-files-dont-move-in-banks>.

⁵Reserve Bank of India, 2016-17, Report on trend and progress in banking.

However, this exposed the firms with high liquidation value to the possibility of liquidation and hence increased the risk of debt financing. Vig (2013) documents that in response to this shock, firms with a high level of tangible assets significantly reduced their debt and specifically secured debt.

I study the difference in response of firms borrowing exclusively from GOBs using the difference-in-difference methodology. Intuitively if liquidation risk is lower for firms borrowing exclusively from GOBs, they will do relatively less reduction in secured debt and the total debt in response to the shock. I find that the decrease in both total debt and secured debt usage is around 2-2.5 percentage points less for GOB firms. These results suggest that firms perceive borrowing from GOBs to be relatively less risky.

Cross-section analysis shows that GOB firms' differential response relative to other firms is stronger in the subsample of firms that are not affiliated to the business groups, are smaller in size, have lower profitability, or have lower interest coverage ratios. In other words, the effect is more substantial among the firms that face higher liquidation risk.

These results imply that continuation incentive coming from other objectives of GOBs seems to dominate the difficulties that GOB managers face in debt restructuring due to fear of investigation for corruption. However, the results are consistent with both incentives. The apparent inconsistency of these results with the second incentive comes from the fact that we typically think that there are only two options while dealing with borrower's default. That is either to liquidate or to reorganize. But there is a third option. Wait and pray that borrower's prospects improve, and it repays the debt in full. There is some evidence that this is the option that GOBs in India seem to have adopted⁶.

Next, I study whether the change in debt usage has any spillover effects on the firms' real investments. It depends on whether firms can substitute a reduction in debt funding from other sources such as retained earnings, dividend cuts, or raising equity. I find that the

⁶Rebello, Joel. "No penalty for not taking decisions, but sword for taking one: why files don't move in banks" Economic Times, July 18, 2019. Accessed November 11, 2019. <https://prime.economictimes.indiatimes.com/news/70269488/fintech-and-bfsi/no-penalty-for-not-taking-decision-but-sword-for-taking-one-why-files-dont-move-in-banks>.

investment rate of GOB firms increases by around 2-3 percentage points relative to other firms. This effect is economically significant, given a mean investment rate of 7.4% in the sample.

I subsequently examine competing explanation for my results and perform different falsification tests. Falsification tests suggest that the results are unlikely to be driven by other confounding factors. Furthermore, the results are robust to various control strategies and continue to hold in different subsamples.

The rest of the paper is organized as follows. I present the theoretical background and related literature in Section 2. In Section 3, I discuss the role of GOBs in India and the institutional environment of the SARFAESI Act. In Section 4, I summarize the data and present the results in Section 5. I address alternate explanations and present results of falsification tests in Section 6. I conclude in Section 7.

2. Conceptual Framework & Related Literature

2.1. *Conceptual Framework*

It is now well recognized that creditors' rights affect both supply and demand for debt. In the standard models (Tirole (2010)), the strengthening of creditors' rights increases the firm's liquidation value. That, in turn, increases pledge-able income and the lender's willingness to supply capital. Furthermore, an increase in liquidation value could reduce the cost of borrowing. For example, in Rodano, Serrano-Velarde, and Tarantino (2016) increase in liquidation value reduces interest cost and, therefore, increases investment. Overall, improvement in creditors' rights would increase the supply of debt while simultaneously decreasing its cost.

However, creditors' rights also affect the demand for debt, which is the focus of this paper. As formulated by Acharya et al. (2011b), the non-linearity of claims of debt and equity holders create a risk of inefficient liquidations in a creditor friendly bankruptcy code. Since

the creditors do not enjoy the upside in the profits, they have a bias towards liquidation. They might liquidate firms that have higher but uncertain continuation value. While the equity holders, being residual claimants, have continuation bias. In a creditor friendly bankruptcy system, creditors control the bankruptcy process. That increases the liquidation risk for the firms that have high liquidation value. Anticipating this, firms use lower debt in creditor friendly countries.

In a two period model, Vig (2013) studies a firm's choice between secured and unsecured debt. In the model, unsecured debt has lower liquidation value and higher cost compared to secured debt, and the owner-manager enjoys private benefit from continuing the firm. However, before the securitization reform, liquidation value for secured debt is also low. Therefore, the creditors prefer continuation over liquidation in dealing with a firm's financial distress. And firms prefer secured debt over unsecured debt due to the lower cost of the former. Post securitization reform, the liquidation value for secured debt increases, and so its cost decreases. However, an increase in liquidation value creates liquidation bias. The loss in private benefit caused by liquidation bias outweighs the benefits due to a reduction in the cost of debt. Hence, the firm reduces the use of secured debt after the reform.

However, in Acharya et al. (2011b) and Vig (2013), the lender has only profit maximization objective, which creates liquidation bias. Yet, GOBs have other objectives apart from profit maximization.

In the literature, there are three views of the government ownership of enterprises. First, the social view argues that the government intervenes in the market through government owned enterprises to address market failures (Stiglitz (1993)). Therefore, they improve social welfare. The second view, the agency view, agrees that government owned enterprises are created to maximize social welfare. Still, it argues that government-owned enterprises' maximization of multiple non-measurable objectives creates governance problems (Banerjee (1997)). Therefore, it leads to corruption and the inefficient allocation of funds. Third, the political view suggests that government owned enterprises are like any other tool for

politicians to seek their political goals (Shleifer (1998)). They use it to exchange political favors or to increase employment before the election.

All three theories agree that GOBs are unlikely to have only a profit maximization objective. They might have a welfare objective such as providing funds to credit rationed borrowers and maximizing economic growth, or they could be serving the personal goals of their managers or the politicians. These other objectives of GOBs create the possibility that they have lower liquidation bias. For example, under the social view, GOBs care about social welfare so they will not force inefficient liquidations. Under the agency view, the GOB managers might not liquidate the firm even though it's optimal for the bank because of the manager's other benefits such as future job prospects in the private sector. While under the political view, GOBs would not liquidate firms for political reasons such as employment preservation. This leads to my main hypothesis that *firms borrowing exclusively from GOBs will have lower liquidation risk*. I test this hypothesis using the natural experiment of securitization reform in India that increased liquidation risk for the firms using debt financing (Vig (2013)).

Next, liquidation bias is a function of two things: the likelihood of the firm facing financial difficulties. Given the firm's financial difficulties, the probability of lender choosing to liquidate the firm. Therefore, if the reduction of secured debt is due to liquidation risk, then the difference in response of GOB firms is likely to be higher in the subsample of firms that are more likely to find themselves in the financial difficulties and do not have access to other sources for support. Therefore, my second hypothesis is that *in cross-section, the difference in GOB firms' response compared to other firms will be higher for firms that are more likely to face financial distress*.

Lastly, in India, public debt markets are not well developed. Therefore, if firms cannot substitute debt financing with retained earnings or equity issues, reduction in debt could have spillover effects on the investments. My third hypothesis is that *post-reform investments of GOB firms would increase compared to other firms*.

2.2. *Related Empirical Literature*

This paper is related to two strands of literature. First, it is associated with the large empirical literature on the impact of creditors' rights on debt financing. There is considerable evidence that stronger creditors rights and efficient contract enforcement have positive effect on the supply of credit (Bae and Goyal (2009); Rodano et al. (2016); Brockman and Unlu (2009); Kyröläinen, Tan, and Karjalainen (2013); Benmelech and Bergman (2011)). For example, Djankov, McLiesh, and Shleifer (2007) find that better creditors' rights are associated with a higher level of aggregate private credit. Haselmann, Pistor, and Vig (2010) show that banks increase loan supply in response to the improvement in creditors' rights in Central and Eastern Europe economies. Djankov, Hart, McLiesh, and Shleifer (2008) find that measure of the effectiveness of the debt contract enforcement mechanism predicts debt market development. Using loan level data, Qian and Strahan (2007) document that stronger creditors' rights are associated with longer debt maturity and lower interest costs.

However, there are set of recent papers that studies the impact of creditors rights on firms' investment (Acharya and Subramanian (2009); Acharya, Amihud, and Litov (2011a); Favara, Morellec, Schroth, and Valta (2017)) and it's demand for debt (Acharya et al. (2011b); Vig (2013); Cho, El Ghoul, Guedhami, and Suh (2014)). This paper is related to them and focuses on the firms' demand for debt.

Acharya et al. (2011b) compare leverage of firms in stronger creditor rights country with that of firms in the weaker creditor rights country. They find that firms with high liquidation value in stronger creditors rights country uses lower leverage to reduce the probability of inefficient liquidation. Consistent with the demand side view, Cho et al. (2014) find that stronger creditor rights are related to lower long term leverage ratios, and firms are less likely to issue long term debt to finance investments.

Vig (2013) is the paper closest to this paper. He studies the effect of the same securitization reform (i.e., SARFAESI Act) in India on the firm's debt structure. Using the reform as a natural experiment that increased the power of secured creditors, he finds that

in response to it, firms in top tercile of tangible assets significantly reduced usage of secured debt compared to firms in the bottom tercile.

Like Vig (2013), I use the SARFAESI Act as a natural experiment. However, I focus on whether liquidation risk of firms borrowing exclusively from GOBs increased by smaller magnitude compared to other firms. To this end, I examine the changes in secured debt usage of GOB firms with that of other firms. To further test the hypothesis that firms borrowing exclusively from GOBs have lower liquidation risk, I analyze differential response as a function of proxies for financial distress. I also examine the spillover effect of differential reduction in secured debt usage on firms' real investments.

Second, this paper is related to growing empirical literature on effect of government owned banks on borrowing firms (Cull and Xu (2003); Firth, Lin, and Wong (2008); Qian and Yeung (2015); Cull et al. (2015); Lin, Srinivasan, and Yamada (2017)). One of the important results in this literature is that firms borrowing from GOBs have lower financial constraints. For example, Lin et al. (2017) find that increase in lending by GOBs during Japan's 1990s financial crisis was positively associated with investment and reduced firms' investment cash-flow sensitivity. State-owned banks also dominate China's banking system. Cull et al. (2015) find that politically connected firms in China have lower financial constraints. While Firth et al. (2008) document that in China negative relation between leverage and investment is less pronounced for low growth firms, firms with poor operating performance, or higher state ownership. In the Indian context, Srinivasan and Thampy (2017) find that firms borrowing exclusively from GOBs have lower investment cashflow sensitivity and higher responsiveness of investment to Tobin's Q. I add to this literature by examining the effect of borrowing from GOBs on firm's liquidation risk.

3. Government Ownership of Banks in India & Background of SARFAESI Act

3.1. Government Ownership of Banks in India

The Indian financial system is bank based. The following data on the mobilization of savings through the different channels in the financial year 2001-02 tells us the importance of the banking sector vis-a-vis capital markets in India⁷. Total credit by commercial banks in 2001-02, was INR 6371.30 Billion⁸, which is significantly higher than funds raised through all other sources combined.

Source	Amount (Billion INR)
Commercial Bank Credit	6371.30
Public Debt Issue: Total	247.71
Public Debt Issue: Non-Financial Firms	82.03
Equity Issue: Total	21.27
Equity Issue: Non-Financial Firms	12.10

After the independence government of India did a series of nationalization of big private banks and brought them under government ownership. Due to these bank nationalizations, the banking sector in India has been historically dominated by government owned banks. However, starting with economic reforms of 1990, new private banks have come up, and foreign banks have also started operating in India. Therefore, their share in total credit has increased to some extent. In 2001-02, government-owned banks' share in the total credit was 74.4% while that of private and foreign banks was 18% and 7.6%, respectively⁹.

⁷For bond and equity issue, SDC Platinum data are used.

⁸Reserve Bank of India, 2001-02, Report on trend and progress in banking

⁹Reserve Bank of India, 2001-02, Report on trend and progress in banking.

The central bank, Reserve Bank of India, classifies banks into four groups based on bank ownership. These four groups are State Bank of India and its associates (GOBs), other nationalized banks (GOBs), private banks, and foreign banks. State Bank of India and its associates and other nationalized banks are owned by the government. Concerning borrower profiles of these banks, Berger et al. (2008) find that firms banking with GOBs in India are less likely to have multiple bank relationships. While firms banking with foreign banks are more likely to have multiple banking relationships, and they are also more likely to have relationships with banks in different ownership groups. Furthermore, they find that large, listed, group firms and foreign firms are more likely to bank with foreign banks.

3.2. Background of SARFAESI Act

Till the early 1990s, the debt contract enforcement process in India was very inefficient. It was filled with long delays in the judicial process and dismal recovery rates. To speed up the legal process of debt recovery, India's government took a series of steps. Two of them were significant. First is the establishment of debt recovery tribunals (DRTs). The law was enacted in 1993. These tribunals were meant to be specialized tribunals for the recovery of a debt due to banks and financial institutions. Second is the enactment of the Securitisation and Reconstruction of Financial Assets and Enforcement of Security Interest (SARFAESI) Act in 2002. Under the SARFAESI Act, banks and financial institutions were provided sweeping powers to recover the secured loans. Before this reform, secured lenders could not take over the collateral of defaulting borrowers to collect their debt. Now under the SARFAESI Act, the banks and financial institutions can directly seize the collateral without a court proceeding. The act requires only two conditions to be met. The loan should have been in default for more than six months, and the lender should give 60 days' notice after the default. Moreover, the borrowers were required to deposit 75% of the claim (and later reduced to 25%) amount borrowed to appeal against the court orders.

There is plenty of evidence that the SARFAESI Act significantly increased the rights of

creditors. Banks' non performing loans (NPAs)¹⁰ and number of cases in DRT fell and debt recovery rates improved¹¹. The corporate lobbies expressed serious concerns about excessive powers given to the creditors. It was perceived that this could lead to excessive liquidations. Further, in response to this reform, firms reduced their secured debt usage (Vig (2013)). It is this shock to liquidation risk that is used as a source of exogenous variation in this paper¹².

4. Data and Summary Statistics

I obtain the data for this study from CMIE-Prowess. It has been used in many papers studying Indian firms, including Vig (2013), Siegel and Choudhury (2012) and Gopalan, Mukherjee, and Singh (2016). It has yearly information of balance sheet and income statement items along with other useful information such as firm's industry category, year of incorporation, and group affiliation. The balance sheet data has a detailed breakup of the firm's borrowings, which helps in identifying the proportion of secured and unsecured debt. Prowess database contains information for both listed and not listed companies. It is unique in one respect that it provides details of all the bankers that the firm has a relationship within a financial year. Following Srinivasan and Thampy (2017), I categorize all the banks that a given firm has a relationship with into three groups: government owned banks (GOBs), private banks, and foreign banks. This provides me with data on the firms' financial information and bankers by the bank ownership group. Using this, I classify firms into two groups: GOB firms and Other firms. GOB firms are firms that have an exclusive relationship with GOBs. Other firms are firms that have an exclusive relationship with private or foreign banks, firms that have a relationship with both private and foreign banks, and firms that have a relationship with GOBs along with private and/or foreign banks. Using data from

¹⁰I have plotted gross NPA as a fraction of gross advances and net NPA as a fraction of net advances between 1999 to 2008 in Appendix Figure A1 left and right panel, respectively. That show that both gross NPA and net NPA fell significantly after the enactment of SARFAESI (i.e., after 2002).

¹¹Under SARFAESI Act, recovery rates till 2008 were 61%. But has fallen to 21.9% in 2013 (Sengupta, Sharma, and Thomas (2016))

¹²Vig (2013) has more detailed discussion of SARFAESI Act and Sengupta et al. (2016) have good summary of evolution of bankruptcy framework in India.

Srinivasan and Thampy (2017), I map prowess NIC product codes to Fama-French industry classifications.

My sample period is 1997 to 2008. I remove financial firms, government owned non-financial firms, and firm-years with negative net worth or sales. Furthermore, there is some evidence that Zombie lending was taking place in India before the reform (Kulkarni (2019)). To rule out the possibility that Zombie lending by GOBs driving differential response, I remove Zombie firms from the sample.

Kulkarni (2019) adopts Caballero, Hoshi, and Kashyap’s 2008 definition of Zombie firm in the Indian context. She defines a firm to be a Zombie firm if it satisfies the following four criteria:

1. Interest coverage ratio (ICR) less than equal to 1.
2. Leverage (total debt to total assets) of greater than 0.20.
3. Taken additional loan in year t .
4. Borrowed funds at an interest rate below prevailing prime lending rates.

I use this definition to classify Zombie firms. However, data on interest rates are not available. I take a conservative approach and classify a firm as a Zombie firm if it satisfies the first three criteria in the year before the reform¹³.

My full sample contains financial and banking relationship information for 10,951 firm-years¹⁴. Due to missing data on some of the variables, the sample size differs in some analyses. The details of the variables used in the study are provided in Appendix Table A1. The data on banking variables are taken from the publicly available web database of Reserve Bank of India (RBI).

Following Vig (2013), I also code year starting from 2002 as the post reform period. The SARFAESI Act came into effect in June 2002. But the discussion on the bill in the

¹³Zombie firms’ consist of around 6% of the observations and the results are robust to including them (Appendix Table A9). Further, the results are similar if I take a more conservative approach and classify all the firms with interest cost (i.e., $\frac{Interest\ Expense}{Avg.\ Borrowings}$) less than the prime lending rate of State Bank of India, the largest bank in India, as Zombie firms (Appendix Table A10).

¹⁴Appendix Table A11 gives details of my sample relative to Vig’s 2013 sample.

parliament was already started by June 2001. Since under my hypothesis, ex-ante effects are more important, I code first financial year ending following June 2001 as the first year after the act¹⁵. I end the sample in the financial year 2008 since the global financial crisis significantly affected the Indian financial sector starting from 2008, and its effects were different for GOBs and other banks (Acharya and Kulkarni (2012)). I have extended the sample until six years after the reform since it takes time for firms to learn that GOBs are not using their powers as stringently as other banks do, and thus to change their debt structure accordingly.

In Table 1, I present summary statistics of the variables used in the analysis. Since the controls are used in lags, I present them in lag in the summary statistics. All variables are winsorized at the 1% and 99% to reduce the influence of outliers. There is considerable variation in all the variables of interest. The mean of *Secured Debt/Total Assets* is 26%, while its standard deviation is 16.1%. The average *Debt/Total Assets* is 32.7% with a standard deviation of 17.9%. These figures highlight that almost 80% of debt is secured debt. Therefore, a very high proportion of debt will be affected by the reform. Finally, profitability, as measured by the ratio of PBITDA to total assets, averages around 12.7%.

5. Empirical Analysis

SARFAESI Act significantly increased the powers of secured lenders. That, in turn, increased the probability of liquidations for firms with high liquidation value. I consider this as an event that increased liquidation risk. To learn whether firms borrowing from GOBs have less risk, I compare the response of GOB firms with that of Other firms using the difference-in-difference methodology. This methodology is motivated by the following argument. If liquidations risk is lower for firms borrowing exclusively from GOBs, they will make relatively less reduction in secured debt and the total debt in response to the

¹⁵Table A7 in Appendix shows that results are robust to the coding year 2003 onwards as post reform period.

shock. But, this is not standard difference-in-difference set up. Since the SARFAESI Act applies to all the firms, I cannot categorize firms strictly into the treatment and control groups. However, based on the above economic argument, firms are classified into effective treatment and control groups. Due to the incentives created by other objectives of GOBs, they are less likely to liquidate than non-GOBs.

As mentioned earlier, Other firms include firms that have an exclusive relationship with private or foreign banks, or with both private and foreign banks or have a relationship with GOBs along with private and/or foreign banks. Ideally, we would like to compare firms that borrow exclusively from GOBs with the firms that borrow only from private banks, domestic or foreign. But the firms that borrow exclusively from private and/or foreign banks are less than 10% in the sample. However, all Other firms are borrowing from at least one privately owned bank. Therefore, the incentive that drives liquidations will be present because the private banker can always push for liquidation even if GOBs lending to the same borrower might be ready for debt restructuring and other arrangements.

My tests will correctly identify the effect of exclusive relationship with GOBs if GOB firms are not significantly different from the Other firms on the aspects that affect liquidation risk. Table 2 presents the mean and standard deviation of various firm characteristics for sample divided by firms that have an exclusive relationship with GOBs in the year before the reform (i.e., the financial year 2001) and firms that did not. We can see that GOB firms are, on average smaller and have a small number of banking relationships than Other firms in the sample. But their financing structure is not significantly different from Other firms.

5.1. *Differential Response of GOB Firms*

In this subsection, I present results for my first hypothesis that GOB firms have less liquidation risk. To this end, I compare the post-reform change in *Secured Debt/Total Assets* of GOB firms to that of Other firms. Intuitively, if GOB firms have lower liquidation risk, they should make a relatively smaller reduction in *Secured Debt/Total Assets* after the reform.

5.1.1. Parallel Trend

I begin by univariate evidence of the differential response of GOB firms to the reform. The dependent variable of interest is *Secured Debt/Total Assets*. Figure 2 left panel presents the mean values for the period of 1997 to 2008. The orange line indicates the average value for GOB firms, and the grey line indicates the average for Other firms. Given earlier arguments, we expect to see two patterns in the plot to reflect the differential response to the shock: first, roughly parallel trends in both groups before the reform that is before 2002. Second, relatively less reduction in *Secured Debt/Total Assets* for GOB firms compared to Other firms after the reform. In the pre-reform period, there is a roughly parallel trend in both groups. Post-reform, *Secured Debt/Total Assets* fall comparatively less for GOB firms.

To further examine the parallel trends assumption, I estimate the following non-parametric version of the difference-in-difference specification:

$$Y_{ijt} = \alpha_i + \delta_{jt} + \sum_{Year=1997}^{2008} \theta_{Year} Year * GOB_i + u_{ijt} \quad (1)$$

here Y_{ijt} is the dependent variable of interest for firm i , in industry j at time t . α_i and δ_{jt} are firm and industry-year fixed effects respectively; $Post_t$ indicates whether firm year belongs to before reform (i.e., years 1997 to 2001) or after reform (i.e., years 2002 to 2008) period; GOB_i indicates whether the firm has an exclusive relationship with GOB in the financial year 2001 or not. The firm fixed effects control for time-invariant firm specific factors affecting the dependent variable while the industry year fixed effects is a non-parametric way of accounting for time varying industry-specific shocks. It controls for both economy wide factors and specific industry shocks in a given year that affects both GOB and Other firms. θ_{Year} are the coefficients of interest. I normalize all the coefficients with respect to the coefficient of the year 2001, i.e. the year immediately preceding the reform and plot them in Figure 2, right panel. Error bars show 95% confidence interval. None of the coefficients for pre-reform years are significant. Therefore, the null hypothesis of parallel trends can not be

rejected.

5.1.2. Matching Estimator

The main results are best captured in Table 3. It presents estimate of average treatment effect on treated (ATT) for *Secured Debt/Total Assets* using Abadie and Imbens (2002) bias corrected matching estimator.

In summary statistics, we noticed some differences between the characteristics of GOB firms from Other firms. It raises the possibility that GOB firms' differential response might be due to the differences in the characteristics of GOB firms from Other firms. To control for these differences, I match Other firms to GOB firms on the basis of pre-reform values of *Log Total Assets*, *Profitability*, *Tangibility*, *Z-Score*, and Fama-French industry code.

However, GOB firms and Other firms are also significantly different in terms of the number of banking relationships. GOB firms are firms that borrow exclusively from GOBs; by definition, they have a small number of banking relationships than Other firms. Average *# Banks* for GOB firms is 2.08 while that for Other firms is 4.73. Since there is limited overlap in *# Banks* between GOB firms and Other firms, I do not include *# Banks* in the matching covariates. However, to remove the bias due to the difference in average *# Banks* between GOB firms and Other firms, I include *# Banks* as covariate in bias adjustment¹⁶.

I use two matching strategies, and for each strategy, I use a 1-to-1 match, and 1-to-4 match¹⁷. In the first strategy, full matching, I use all the firms from Other group in the sample to match with GOB firms. We noted earlier that Other firms have higher *# Banks* than GOB firms. Therefore, in the second strategy, restricted matching, I restrict control group to Other firms that have *# Banks* less than or equal to maximum *# Banks* among GOB firms (which is 9). This approach reduces the difference in *# Banks* between GOB firms and Other firms. Table 3 Panel A gives results for full matching while Panel B gives

¹⁶In Abadie and Imbens (2002) matching estimator, we can include covariate into bias adjustment even if it is not included in matching.

¹⁷1-to-1 match is a natural starting point. However, it is not statistically efficient. 1-to-4 match gives a good balance between the bias and the precision of the matching estimator(Abadie and Imbens (2011)).

that for restricted matching¹⁸.

In Panel A, after the reform, Other firms reduce their *Secured Debt/Total Assets* by 2.7 percentage points while GOB firms reduce their *Secured Debt/Total Assets* by only 40 basis points. Bias corrected estimate of the relative increase in *Secured Debt/Total Assets* of GOB firms is 2.4 and 2.3 percentage points for 1-to-1 and 1-to-4 match, respectively. For restricted matching in Panel B, the relative increase in *Secured Debt/Total Assets* of GOB firms is 2.3 and 2.1 percentage points for 1-to-1 and 1-to-4 match, respectively.

5.1.3. Regression Estimate

To further evaluate the differential response to the shock, I estimate the following difference-in-difference regression equation:

$$Y_{ijt} = \alpha_i + \delta_{jt} + \theta Post_t * GOB_i + \omega X_{ijt} + u_{ijt} \quad (2)$$

here X_{ijt} are control variables. I control for standard capital structure control variables (Rajan and Zingales (1995)) except for TobinQ. I exclude TobinQ due to measurement error concerns (Erickson and Whited (2000))^{19,20}. To address within firm correlation in errors, I have clustered the standard errors at the firm level (Bertrand, Duflo, and Mullainathan (2004)). I focus on the coefficient θ . It captures the differential response of GOB firms compared to Other firms to the shock. The results are presented in Table 4. In Column 1, I report the results without controls and fixed effects. I add controls in Column 2, and firm and industry year fixed effects in Column 3 and 4, respectively. We can see that *Secured Debt/Total Assets* goes up by 2 percentage points in Column 1 and 1.9 percentage points in Column 2 for GOB firms compared to Other firms in the sample. The effect is of similar magnitude after adding firm and industry year fixed effects in Column 3 and 4. In Column

¹⁸No. of firms in the matching estimates are lower than that in the full sample since matching estimator requires data in both the pre and the post-reform period.

¹⁹Table A8 in appendix shows that results do not change due to the inclusion of TobinQ as a control.

²⁰In this analysis, I have used Rajan and Zingales (1995) controls in lags. However, the results are robust to including controls in level. They are given in Appendix Table A6.

5, I add interaction of controls with *Post*. The coefficient θ is still 2 percentage points. The coefficient marginally increases after controlling for *Z-Score* (Altman (2005)) and *# Banks* in Column 6 and 7, respectively.

A problem with difference-in-difference methodology is that serial correlation in errors at the group level can lead to bias in estimated standard errors and result in the overestimation of statistical significance (Bertrand et al. (2004)). To alleviate this concern, I perform a non-parametric Fisher’s exact test for $\theta = 0$ (Chetty, Looney, and Kroft (2009)). To get a “placebo effect”, first, I randomly assign firms to GOB and Other categories such that the proportion of GOB firms is the same as actual in my sample (i.e., 55%). I then estimate Equation (2) to calculate θ , assuming that this placebo assignment is the actual assignment. Fisher’s exact test requires that I should repeat this procedure for all possible permutation of GOB and Other firm groupings. But due to computational difficulties, I repeat it for 10000 times. It would give us statistically similar results. Next, I calculate the empirical cumulative distribution function (CDF), $\hat{F}_n(x)$, for these effects. This empirical CDF, $\hat{F}_n(x)$, gives a p-value for the null hypothesis that $\theta = 0$. Intuitively, if there were a differential response to the shock, we would expect the estimated coefficient to be in the right tail of calculated placebo effects. This test does not suffer from the overestimation of the statistical significance problem of the t-test because it does not assume a particular error structure.

In Figure 3, I present the results of Fisher’s exact test for *Secured Debt/Total Assets*. The figure shows the empirical CDF of calculated placebo effects $\hat{F}_n(x)$. The vertical line in the figure represents the differential effect documented in Table 4 Column 4. The value of empirical distribution function is $\hat{F}_n(0.027) = 0.9997$. This p-value is lower than that obtained for the t-test.

Overall, these results suggest that GOB firms reduced *Secured Debt/Total Assets* by about 2-2.5 percentage points less compared to Other firms in response to the shock. Further, analysis of total debt usage suggests that firms do not seem to substitute reduction in secured debt by borrowing on an unsecured basis. The results for total debt are given in Appendix

Table A4. *Debt/Total Assets* reduces by roughly 2-2.5 percentage points

In the introduction, we noted that GOBs in India have potentially two opposing incentives for dealing with the default of borrowers. First, GOBs, due to their objectives beyond profit maximization, are likely to choose continuation over liquidation when dealing with borrowers' financial difficulties. However, to continue a firm facing temporary financial problems might involve debt resurrecting and some debt forgiveness. This reorganization gives rise to an opposing force. In India, GOB managers are considered as a public servant. And their banking decisions, such as debt restructuring, can be investigated for corruption by law enforcement agencies. The fear of investigation could make reorganization difficult.

The results presented so far would imply that the former incentive seems to dominate the latter. However, the results are consistent with both incentives. The apparent inconsistency of these results with the second incentive comes from the fact that we typically see only two options while dealing with borrowers' default. That is either to liquidate or to reorganize. However, a third option is to wait for borrowers' prospects to improve, and it repays the debt in full. There is some evidence that this is the option that GOBs in India seem to have adopted. For example, Papia Sengupta, Executive Director of Bank of Baroda, third-largest bank in India²¹ commented in a recent interview²²

“After these issues(the investigations), bankers have been super cautions. Decisions will not be taken, and this is worse than taking a wrong decision”

They [one or two banks in a bankers consortium] will try to delay the matter by either seeking revaluation, asking for a higher amount in case it's sale, or just raising more questions. Unfortunately, today there is no penalty for not taking a decision, but you could be punished for taking one”

²¹ “Bank of Baroda becomes second-largest PSU, third-largest bank in India” Business Today, April 1, 2019. Accessed November 11, 2019. <https://www.businesstoday.in/top-story/bank-of-baroda-becomes-second-largest-psu-third-largest-bank-in-india/story/332888.html>

²² Rebello, Joel. “No penalty for not taking decisions, but sword for taking one: why files don't move in banks” Economic Times, July 18, 2019. Accessed November 11, 2019. <https://prime.economictimes.indiatimes.com/news/70269488/fintech-and-bfsi/no-penalty-for-not-taking-decision-but-sword-for-taking-one-why-files-dont-move-in-banks>.

Although these observations are recent and my sample period ended in 2008, these incentives are likely to be present during my sample period. It is during the economic slowdown when a high proportion of borrowers face financial difficulties, problems created by weak incentives come to light.

5.2. *Cross-sectional Results*

Now, I present results of my second hypothesis that in cross-section the difference in response of GOB firms compared to Other firms will be higher for firms that are more likely to face financial distress. Since the lender is more likely to liquidate a distressed firm if liquidation value is high, I focus on the subsample of firms with the above median level of tangible assets in the year before the reform for this analysis.

5.2.1. *Profitability & Interest Coverage Ratio*

Firms with lower profitability and lower interest coverage ratio are more likely to find themselves in financial difficulties than the firms with higher profitability and higher interest coverage ratio. Therefore, the differential effect between GOB firms and Other firms should be more substantial in these subsample of firms. To examine the differences in the impact on secured debt based on the profitability, and the interest coverage ratio of the firm, I start with firms with tangible assets above the median in 2001 and divide it into two groups, above and below the median level for each variable²³. I use $PBITDA/Total\ Assets$ and $PBITDA/Interest\ Costs$ as the proxy for profitability and interest coverage ratio, respectively. I report the results in Table 5. Column 1, and Column 3 show results for the subsample of firms with above median level of profitability and interest coverage ratio, respectively, while Column 2, and Column 4 give that for below median. The impact on $Secured\ Debt/Total\ Assets$ is stronger in the subsample of firms with lower profitability, and

²³Since the median is calculated based on the entire Prowess universe, and Prowess has missing information on bank affiliations for many firms, the number of observations above and below median might not be equal in some cases.

lower interest coverage ratio than the subsample of firms with higher profitability and higher interest coverage ratio.

5.2.2. *Small Firms & Non-Group Firms*

During the financial distress, all the firms find it challenging to raise funds, but it would be relatively more challenging for smaller firms. Therefore, small firms are more vulnerable, and they would have a higher liquidation risk. To examine the differences in effect on secured debt based on the firm size, I divide firms into two groups based on median *Total Assets* in the year before the reform. The results are presented in Column 5 and 6. The difference in response between GOB firms and Other firms is substantially higher in the subsample of firms with the below median level of *Total Assets*.

Gopalan, Nanda, and Seru (2007) documented that business groups in India use inter-group loans to transfer funds to support group firms facing financial difficulties. This support helps member firm to avoid default. However, the standalone firm does not have such support in financial distress. And therefore, they are more at risk of default in adverse financial conditions. To examine the differences in effect on secured debt between group firms and stand alone firms, I divide the sample of firms with above median level of tangible assets into the following two groups: standalone firms and group firms. The results are presented in Column 7 and 8. Among standalone firms, post-reform *Secured Debt/Total Assets* of GOB firms increases by 5.1 percentage points relative to Other firms. While among business group firms, the response of GOB firms is not significantly different from Other firms.

Overall, the cross-section results show that the differential effect between GOB firms and Other firms is higher in the subsample of firms that are more likely to face financial difficulties.

5.3. *Spillover effects to the investments*

In emerging markets like India, capital markets are not well developed. And we noted that reduction in secured debt is not substituted with an increase in unsecured debt. Thus, cut in debt usage could affect the real investments of firms. In this subsection, I test my third hypothesis that post reform investments rate of GOB firms would increase compared to Other firms.

5.3.1. *Matching Estimator*

To examine whether the reduction in secured debt spillovers to firms' real investments, I estimate the average treatment effect on treated (ATT) for *Investments* using Abadie and Imbens (2002) bias corrected matching estimator. In this analysis, I follow the same approach as in Section 5.1.2. I match Other firms to GOB firms on the basis of pre-reform values of *Log Total Assets*, *Profitability*, *Tangibility*, *Z-Score*, and Fama-French industry code, and include *# Banks* as additional covariate in bias adjustment. Further, in this section, also I employ full matching and restricted matching strategies with both 1-to-1 and 1-to-4 match.

The results are presented in Table 6. Panel A tabulates results for full matching while the results for restricted matching are tabulated in Panel B. In Panel A, after the reform, Other firms' *Investments* decrease by 3 percentage points. In contrast, GOB firms' *Investments* is roughly unchanged. Bias corrected estimate of the relative increase in *Investments* of GOB firms is 2.3 and 3.2 percentage points for 1-to-1 and 1-to-4 match, respectively. For restricted matching in Panel B, the relative increase in *Investments* of GOB firms is 2.4 and 3.1 percentage points for 1-to-1 and 1-to-4 match, respectively. The relative increase in GOB firms *Investments* is economically significant, given the mean investment rate of 7.4% in the sample.

5.3.2. Regression Estimate

To further evaluate the spillover effect of the shock to the investments, I estimate the Equation (2) taking *Investments* as the dependent variable. I use the standard determinant of investments as controls except for TobinQ²⁴. The controls include cashflow from operation scaled by lag of total assets, log of total assets, and lag of leverage (Srinivasan and Thampy (2017)).

The results for Equation (2) are reported in Table 7. The results are statistically significant for both usual t-test and non-parametric Fisher’s exact test²⁵. Column 1 reports results without controls and fixed effects. Controls are added in Column 2, while firms and industry year fixed effects are added in Column 3 and 4, respectively. *Investments* for GOB firms increased by 2.1 and 2 percentage points in Column 1 and Column 2, respectively. In Column 3 and 4, the effect increases to 2.8 and 2.7 percentage points, respectively. Finally, in Column 5, I add interaction of controls with *Post*. The magnitude of the coefficient θ decreases, but it is still significant. Overall, these results suggest that the reduction in debt usage has a spillover effect on the firm’s investments.

6. Alternative Explanations & Falsification Tests

In this section first I discuss results of falsification tests and then address certain alternative explanations for my results.

6.1. Falsification Tests

It is possible that GOB and Other firms randomly diverge in their usage of secured debt, and I am capturing that effect. In other words, the difference-in-difference effect I have documented is not due to the lower liquidation risk of borrowing from GOBs. To rule out

²⁴Table A8 in Appendix shows that results do not change due to the inclusion of TobinQ as a control.

²⁵ The results for Fisher’s exact test are reported in Figure A4. The procedure used for Fisher’s exact test for the investments is the same as that described for the ratio of secured debt to total assets earlier.

this possibility, I perform three falsification tests. First, I analyse the effect on the use of unsecured debt. The SARFAESI Act applies only to secured debt. So under my hypothesis, the use of unsecured debt should not be affected. I estimate Equation (2) for the ratio of unsecured debt to total assets. The results are presented in Table 8. We can see that the effect for unsecured debt is not significant in any of the specifications.

Second, I perform split-sample analysis of firms with the above and below the median level of tangible assets before the reform. SARFAESI Act gives the power for seizing the collateral to the lender. That, in turn, increases the possibility of liquidation for firms with high liquidation value. Under this hypothesis, the shock will affect more to firms with a high level of tangible assets compared to firms with a low level of tangible assets. Results for the firms with the above and below the median level of tangible assets are given in Table 9, Panel A and B, respectively. The coefficient of $Post * GOB$ is larger and statistically significant at 1% level in above median tangibility sample while it is small and statistically insignificant for the below median tangibility sample.

Lastly, I falsely assign 1999 as the year of the act and the study sample from the year 1995 to 2001. Here, I consider the period from 1995 to 1998 as the pre-reform period and the period of 1999 to 2001 as the post-reform period. I estimate Equation (2) for this sample. The results are tabulated in Panel B. The coefficient for $Post * GOB$ is insignificant and close to zero for all the dependent variables. Overall these falsification tests suggest that my results are unlikely to be the outcome of random divergence in trend between GOB and Other firms.

6.2. *Alternative Explanations*

As noted earlier, SARFAESI Act is likely to affect more to the firms with a higher level of tangible assets. Instead of lower liquidation risk of borrowing from GOB, the pattern I document could also be observed if either only firms with the low level of tangible assets borrow exclusively from GOBs, or these results are driven by a subsample of firms with a low

level of tangible assets. There are two reasons why this is unlikely to be the case. First, the asset tangibility level for GOB firms is not significantly different from Other firms (Table 2). In fact, the average asset tangibility level for GOB firms is slightly higher. Second, as discussed above, for the subsample of firms with tangible assets below the median, the difference in GOB firms' response compared to other firms is smaller and not statistically significant.

The reform was carried out to tackle the problem of rising NPAs of banks and financial institutions. As discussed earlier, after the act, debt and specially secured debt usage have reduced. Another alternative explanation is as follows. Reduction in leverage after the passage of act could be due to the recovery of NPAs by banks using the power vested to them under the act. And therefore, GOB firms' relatively lower reduction in debt could be explained by the fact that NPAs of GOBs were comparatively lower before the reform. So their borrowers have to make relatively less repayment compared to the borrowers of other banks. But the aggregate NPA data does not support this explanation. Figure A1 in Appendix gives the plot of the ratio gross NPAs to gross advances and net NPAs to net advances, respectively, for the period 1999 to 2008. The three lines represent three groups: government owned banks (GOB), privately owned banks (POB), foreign owned banks (FOB). We can see in the figure that for GOBs, the ratio of gross NPAs to gross advances and the ratio of net NPAs to net advances were highest before the reform. It reduces more significantly after the reform and converges with that of other bank ownership groups. This pattern in NPAs suggests that if anything, GOBs have recovered more money using the act's powers.

6.3. Robustness Tests

SARFAESI Act gives more power to banks for their debt recovery. Therefore, it will have more effect on firms that do not have access to bond markets. It is possible that difference in response of GOB firms compared to Other firms is due to their differential access to bond

markets. To rule out this possibility, I estimate Equation (2) after removing firms with access to bond markets²⁶. Further, I examine the robustness of my results to inclusion of Zombie firms, restricting sample to firms surviving in the last year of sample period, and use of alternate proxy for credit risk (distance-to-default²⁷). I report the results of these robustness tests in Table 10. They are similar to the baseline results.

SARFAESI Act was passed in June 2002 while my sample extends till 2008. As argued earlier, the sample covers post act data for six years since it takes time for firms to learn that GOBs are not using their powers as stringently as other banks. But it raises natural concern that my results might be picking up effect that occurs far after the enactment of SARFAESI, and the reform does not drive it. To address this concern, I restrict the sample to 2004 and present the results in Appendix Table A7 Panel A. We can see that effect is still present, although it gets weaker both economically and statistically. Reduction in statistical significance is likely to be driven by a decrease in sample size. Further, Panel B and C show that the results are robust to considering 2003 as the year of reform and excluding the firms that switch GOB categorization after the reform.

7. Conclusion

GOBs have other explicit or implicit objectives apart from profit maximization. Do the incentives created by these objectives affect the liquidation risk of firms borrowing from GOBs? It is an important question given a significant number of firms borrow from GOBs in emerging markets. In this paper, I have studied this question using a natural experiment of securitization reform in India. The reform increased the probability of liquidation for firms and led to an increase in the liquidation risk. In response to this shock, firms reduced their usage of secured debt and total debt.

²⁶To identify firms with access to bond markets, I have matched Prowess firms with SDC Platinum bond issue data using Fuzzy name matching techniques.

²⁷The number of observations for distance-to-default is lower due to unavailability daily stock price data to calculate stock volatility.

I find that GOB firms reduce their secured debt and total debt usage relatively less than other firms. In the cross-section, the differential response is stronger in firms that are more likely to face financial difficulties and firms that will find it more challenging to get support from other sources. This reduction in secured debt had a significant spillover effect on their real investments. The investment rate for GOB firms relative to Other firms increases significantly.

Overall, these results are consistent with anecdotal evidence that GOBs as lenders are comparatively less stringent in debt recovery and suggest that firms borrowing exclusively from GOBs have less liquidation risk.

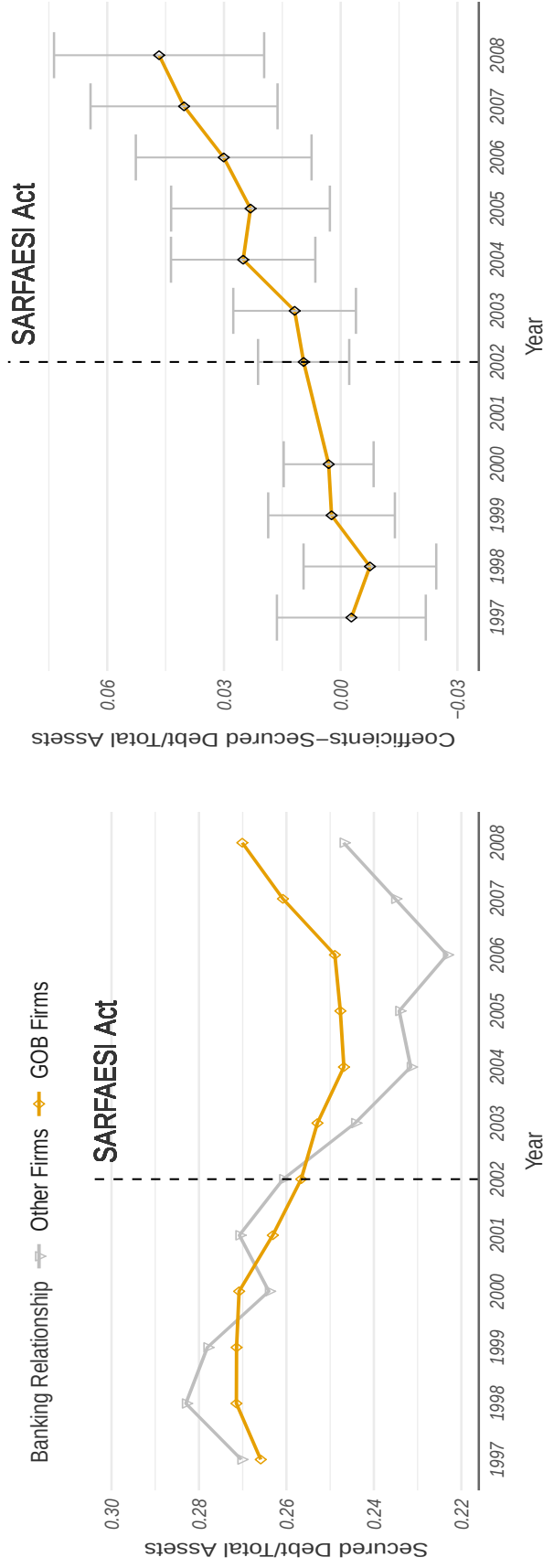
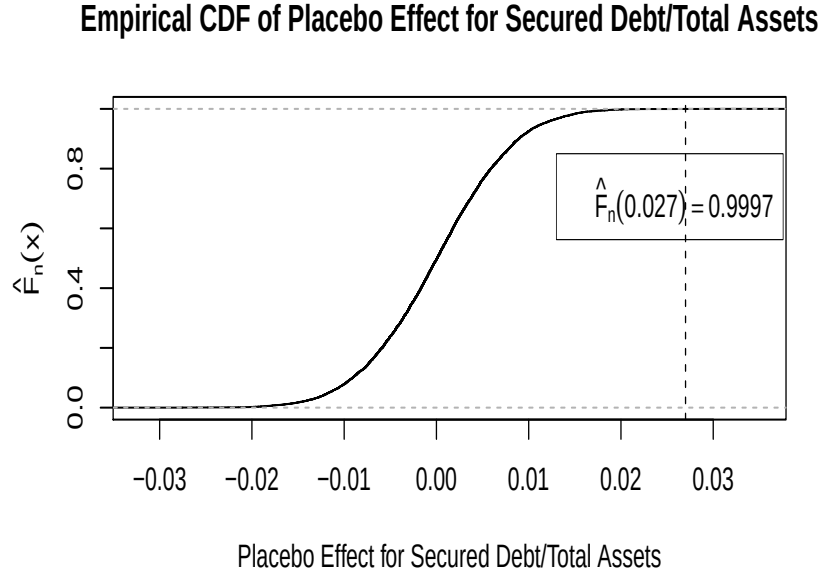


Fig. 1. Dynamic Difference-in-Difference Coefficients for Secured Debt/Total Assets

This figure is for *Secured Debt/Total Assets*. Left panel shows the plot of group wise yearly average for GOB firms and Other firms. GOB firms are firms having exclusive relationship with GOB in the year before the reform. The rest of the firms belong to Other group. Right panel shows the plot of coefficients θ_{Year} for the non-parametric difference-in-difference regression: $Y_{ijt} = \alpha_i + \delta_{jt} + \sum_{Year=1997}^{2008} \theta_{Year} * GOB_i + u_{ijt}$. α_i and δ_{jt} are firm and industry-year fixed effects respectively; $Post_t$ indicates whether firm year belongs to before act (1997 to 2001) or after act (2002 to 2008) period; GOB_i indicates whether the firm has exclusive relationship with GOBs in financial year 2001 or not. The standard errors are clustered at firm level. Error bars show 95% confidence interval.

Fig. 2. Empirical CDF of Placebo Effect for Secured Debt/Total Assets



This figure shows the empirical CDF of placebo effects calculated for *Secured Debt/Total Assets*. The CDF is constructed from 10,000 estimates of differential effect i.e., interaction term using following equation for the baseline sample: $Y_{ijt} = \alpha_i + \delta_{jt} + \theta Post_t * GOB_i + \omega X_{ijt} + u_{ijt}$. No parametric smoothing is used: the empirical CDF appears smooth due to the 10,000 data points used to construct it. The vertical line denotes the estimated effect presented in Table 4 Column 4.

Table 1: Summary Statistics

Variable	n	mean	sd	median	min	max
Debt/Total Assets	10,951	0.327	0.179	0.325	0.001	0.843
Secured Debt/Total Assets	10,951	0.260	0.161	0.248	0.000	0.736
Investments	10,951	0.074	0.128	0.034	-0.243	1.059
Tangibility _{<i>t</i>-1}	10,951	0.387	0.191	0.380	0.004	0.953
Log Sales _{<i>t</i>-1}	10,951	6.144	1.648	6.183	-1.204	11.257
Log Total Assets	10,951	6.503	1.516	6.378	2.241	11.787
Profitability _{<i>t</i>-1}	10,951	0.127	0.076	0.123	-0.163	0.484
# Banks	10,951	3.189	2.890	2.000	1.000	42.000
TobinQ _{<i>t</i>-1}	10,189	1.047	0.748	0.873	0.135	13.144

This table reports summary statistics. All variables are defined in Appendix Table A1. This excludes observations for financial firms and state owned non-financial firms. Sample Period is 1997 to 2008.

Table 2: Summary Statistics By GOB Categorization

Variable	<i>GOB Firms</i>			<i>Other Firms</i>			<i>Difference</i>		
	mean	sd	Obs.	mean	sd	Obs.	diff	se	t-statistic
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Debt/Total Assets	0.323	0.176	6,400	0.333	0.185	4,551	-0.010	0.004	-2.866
Secured Debt/Total Assets	0.262	0.156	6,400	0.257	0.168	4,551	0.004	0.003	1.421
Investments	0.071	0.127	6,400	0.077	0.128	4,551	-0.006	0.002	-2.529
Tangibility _{<i>t</i>-1}	0.398	0.186	6,400	0.371	0.197	4,551	0.028	0.004	7.397
Log Sales _{<i>t</i>-1}	5.759	1.463	6,400	6.686	1.739	4,551	-0.927	0.032	-29.333
Log Total Assets	6.056	1.301	6,400	7.131	1.574	4,551	-1.075	0.028	-37.794
Profitability _{<i>t</i>-1}	0.122	0.077	6,400	0.134	0.074	4,551	-0.012	0.001	-8.188
# Banks	2.088	1.665	6,400	4.737	3.478	4,551	-2.649	0.056	-47.648
TobinQ _{<i>t</i>-1}	0.938	0.566	5,876	1.195	0.920	4,313	-0.257	0.016	-16.199

This table reports summary statistics by GOB relationship of firms. Firms are divided into two groups. The first group is GOB firms. These are firms having exclusive relationship with GOBs in the year before the reform. That is in financial year 2001. The rest of the firms belong to Other group. Sample Period is 1997 to 2008. Difference in average value of variable between GOB and Other firms during the sample period is given in Column 7 and its standard error is given in Column 8.

Table 3: Bias Corrected Matching Estimator—Secured Debt/Total Assets

<i>Panel A: Full Matching</i>	
GOB Firms: After - Before	-0.004
Other Firms: After - Before	-0.027
Bias Corrected Matching Estimate (1 to 1 Match)	0.024** (0.011)
Bias Corrected Matching Estimate (1 to 4 Match)	0.023** (0.010)
No. of Firms	850
<i>Panel B: Restricted Matching— # Banks $\leq$ Max (GOB Firms' #Banks)</i>	
GOB Firms: After - Before	-0.004
Other Firms: After - Before	-0.027
Bias Corrected Matching Estimate (1 to 1 Match)	0.023** (0.011)
Bias Corrected Matching Estimate (1 to 4 Match)	0.021** (0.010)
No. of Firms	836
<i>Note:</i>	*p<0.1; **p<0.05; ***p<0.01

This table presents results of the bias-corrected matching estimator (Abadie and Imbens (2002)) for *Secured Debt/Total Assets* around SARFAESI Act. The estimator minimizes the Mahalanobis distance between the covariates. Panel A presents the results for all firms in the sample. In Panel B matching for the control group is restricted to the firms that have banking relationships less than or equal to maximum among GOB firms. The first row in both panels contains the differences in the outcome variable for GOB firms between the pre-reform (1997-2001) and the post-reform (2002-2008) period, and the second row contains the differences in the outcome variables for matched control group firms over the same period. The third row gives the estimate for 1-to-1 match while the estimates for 1-to-4 match are given in fourth row.

Table 4: Results For Secured Debt/Total Assets

	<i>Dependent variable: Secured Debt/Total Assets</i>						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Log Sales _{t-1}		0.026*** (0.002)	0.011*** (0.003)	0.013*** (0.004)	0.018*** (0.004)	0.011*** (0.004)	0.012*** (0.004)
Tangibility _{t-1}		0.254*** (0.016)	0.056*** (0.019)	0.060*** (0.019)	0.090*** (0.020)	0.037** (0.018)	0.060*** (0.019)
Profitability _{t-1}		-0.303*** (0.034)	-0.282*** (0.028)	-0.287*** (0.029)	-0.206*** (0.034)	-0.140*** (0.027)	-0.285*** (0.029)
Z-Score						-0.026*** (0.002)	
# Banks							0.002 (0.001)
Post*GOB	0.020** (0.008)	0.019** (0.008)	0.025*** (0.008)	0.027*** (0.008)	0.020** (0.008)	0.024*** (0.007)	0.027*** (0.008)
Post*Log Sales _{t-1}					-0.007** (0.003)		
Post*Tangibility _{t-1}					-0.062*** (0.019)		
Post*Profitability _{t-1}					-0.158*** (0.050)		
Post*Z-Score						0.002 (0.002)	
Observations	10,951	10,951	10,951	10,951	10,951	10,951	10,951
Adjusted R ²	0.005	0.140	0.698	0.703	0.706	0.728	0.703
Controls	No	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	No	No	Yes	Yes	Yes	Yes	Yes
Industry Year FE	No	No	No	Yes	Yes	Yes	Yes

Note:

*p<0.1; **p<0.05; ***p<0.01

This table reports results for the regression: $Y_{ijt} = \alpha_i + \delta_{jt} + \theta Post_t * GOB_i + \omega X_{ijt} + u_{ijt}$. Here, Y_{ijt} denotes *Secured Debt/Total Assets* for firm i , in industry j at time t . α_i and δ_{jt} are firm and industry-year fixed effects respectively; $Post_t$ indicates whether firm year belongs to before act (1997 to 2001) or after act (2002-2008) period; GOB_i indicates whether the firm has exclusive relationship with GOB in financial year 2001 or not; X_{ijt} are control variables. The standard errors are clustered at firm level and reported in the brackets. Column 1 reports results without controls and fixed effects. Rajan and Zingales (1995) controls are added in Column 2, while firm and industry year fixed effects are added in Column 3 and 4, respectively. In Column 5, I add interaction of $Post$ with Rajan and Zingales (1995) controls. I control for Z -Score and $\# Banks$ in Column 6 and 7, respectively.

Table 5: Cross Section Results—Secured Debt/Total Assets

Dependent variable: Secured Debt/Total Assets									
	<i>Profitability</i>		<i>Interest Cov. Ratio</i>		<i>Size</i>		<i>Group</i>		
	<i>High</i>	<i>Low</i>	<i>High</i>	<i>Low</i>	<i>Big</i>	<i>Small</i>	<i>Affiliated</i>	<i>Standalone</i>	
Post*GOB	0.009 (0.011)	0.061*** (0.021)	0.015 (0.014)	0.028* (0.014)	0.011 (0.011)	0.069*** (0.020)	−0.003 (0.015)	0.051*** (0.013)	
Total Obs	3,723	1,832	2,620	2,820	3,642	1,913	2,160	3,395	
Treated Obs	2,094	1,308	1,569	1,750	1,889	1,513	1,055	2,347	
Adjusted R ²	0.700	0.738	0.691	0.695	0.727	0.656	0.748	0.700	
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	
Industry Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	

Note:

*p<0.1; **p<0.05; ***p<0.01

This table reports cross section results based on profitability, interest coverage ratio, firm size, and business group affiliation. The value of *PBITDA/Total Assets*, *PBITDA/Interest Costs*, *Total Assets*, and *Group Dummy* in 2001 are used as proxy for profitability, interest coverage ratio, firm size, and business group affiliation, respectively. Median value of the proxy for concerned variable is used to divide the sample into two groups. The following regression equation is used for the analysis: $Y_{ijt} = \alpha_i + \delta_{jt} + \theta Post_t * GOB_i + \omega X_{ijt} + u_{ijt}$. Here, Y_{ijt} denotes the dependent variable, *SecuredDebt/TotalAssets*, for firm i , in industry j at time t . α_i and δ_{jt} are firm and industry-year fixed effects respectively; *Post_t* indicates whether firm year belongs to before act (1997 to 2001) or after act (2002 to 2008) period; *GOB_i* indicates whether the firm has exclusive relationship with GOB in financial year 2001 or not; X_{ijt} are control variables. The coefficient of interest is θ . The standard errors are clustered at firm level and reported in the brackets. The sample is restricted to firms with above median level of tangible assets in the year of prior to reform.

Table 6: Bias Corrected Matching Estimator-Investments

<i>Panel A: Full Matching</i>	
GOB Firms: After - Before	0.001
Other Firms: After - Before	-0.030
Bias Corrected Matching Estimate (1 to 1 Match)	0.023** (0.010)
Bias Corrected Matching Estimate (1 to 4 Match)	0.032*** (0.008)
No. of Firms	850
<i>Panel B: Restricted Matching- # Banks $\leq$ Max (GOB Firms' #Banks)</i>	
GOB Firms: After - Before	0.001
Other Firms: After - Before	-0.030
Bias Corrected Matching Estimate (1 to 1 Match)	0.024** (0.010)
Bias Corrected Matching Estimate (1 to 4 Match)	0.031*** (0.009)
No. of Firms	836
<i>Note:</i>	*p<0.1; **p<0.05; ***p<0.01

This table presents results of the bias-corrected matching estimator (Abadie and Imbens (2002)) for *Investments* around SARFAESI Act. The estimator minimizes the Mahalanobis distance between the covariates. Panel A presents the results for all firms in the sample. In Panel B matching for the control group is restricted to the firms that have banking relationships less than or equal to maximum of among GOB firms. The first row in both panels contains the differences in the outcome variable for GOB firms between the pre-reform (1997-2001) and the post-reform (2002 and 2008) period, and the second row contains the differences in the outcome variables for matched control group firms over the same period. The third row gives the estimate for 1-to-1 match while the estimates for 1-to-4 match are given in fourth row.

Table 7: Results For Investments

	<i>Dependent variable:</i>				
	Investments				
	(1)	(2)	(3)	(4)	(5)
Cashflow		0.148*** (0.021)	0.121*** (0.023)	0.121*** (0.023)	0.066*** (0.032)
Log Total Assets		0.013*** (0.001)	0.058*** (0.005)	0.068*** (0.006)	0.077*** (0.007)
Debt/Total Assets_1		−0.001 (0.009)	−0.129*** (0.019)	−0.123*** (0.019)	−0.133*** (0.023)
Post*GOB	0.021*** (0.005)	0.020*** (0.005)	0.028*** (0.006)	0.027*** (0.006)	0.018** (0.007)
Post*Cashflow					0.094** (0.039)
Post*Log Total Assets					−0.011*** (0.002)
Post*Debt/Total Assets_1					0.027 (0.019)
Observations	10,951	10,951	10,951	10,951	10,951
Adjusted R ²	0.002	0.045	0.184	0.203	0.207
Controls	No	Yes	Yes	Yes	Yes
Firm FE	No	No	Yes	Yes	Yes
Industry Year FE	No	No	No	Yes	Yes

Note:

*p<0.1; **p<0.05; ***p<0.01

This table reports results for the regression: $Y_{ijt} = \alpha_i + \delta_{jt} + \theta Post_t * GOB_i + \omega X_{ijt} + u_{ijt}$. The dependent variable is *Investments*. Here, Y_{ijt} denotes the dependent variable of interest for firm i , in industry j at time t . α_i and δ_{jt} are firm and industry-year fixed effects respectively; $Post_t$ indicates whether firm year belongs to before act (i.e., years 1997 to 2001) or after act (i.e., years 2002 to 2008) period; GOB_i indicates whether the firm has exclusive relationship with GOB in financial year 2001 or not; X_{ijt} are control variables. The coefficient of interest is θ . The standard errors are clustered at firm level and reported in the brackets. Column 1 reports results without controls and fixed effects. Controls are added in Column 2, while firm and industry year fixed effects are added in Column 3 and 4, respectively. In Column 5, I add interaction of controls with *Post*.

Table 8: Results For Unsecured Debt/Total Assets

	<i>Dependent variable: Unsecured Debt/Total Assets</i>			
	(1)	(2)	(3)	(4)
Post*GOB	-0.00003 (0.005)	0.0005 (0.005)	0.001 (0.005)	0.001 (0.005)
Observations	8,809	8,809	8,809	8,809
Adjusted R ²	0.005	0.020	0.498	0.509
Controls	No	Yes	Yes	Yes
Firm FE	No	No	Yes	Yes
Industry Year FE	No	No	No	Yes

Note:

*p<0.1; **p<0.05; ***p<0.01

This table reports results for the regression: $Y_{ijt} = \alpha_i + \delta_{jt} + \theta Post_t * GOB_i + \omega X_{ijt} + u_{ijt}$. The dependent variable is *UnsecuredDebt/TotalAssets*. Here, Y_{ijt} denotes the dependent variable of interest for firm i , in industry j at time t . α_i and δ_{jt} are firm and industry-year fixed effects respectively; $Post_t$ indicates whether firm year belongs to before act (i.e., years 1997 to 2001) or after act (i.e., years 2002 to 2008) period; GOB_i indicates whether the firm has exclusive relationship with GOB in financial year 2001 or not; X_{ijt} are control variables. The coefficient of interest is θ . The standard errors are clustered at firm level and reported in the brackets. Column 1 reports results without controls and fixed effects. Controls are added in Column 2, while firm and industry year fixed effects are added in Column 3 and 4, respectively.

Table 9: Falsification Tests

	<i>Dependent variable:</i>					
	Debt/Total Assets	Secured Debt/Total Assets	Investments			
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A–Below Median Tangibility Sample</i>						
Post*GOB	0.012 (0.014)	0.007 (0.013)	0.014 (0.012)	0.009 (0.011)	−0.001 (0.008)	0.004 (0.008)
Observations	3,733	3,733	3,733	3,733	3,733	3,733
Adjusted R ²	0.676	0.707	0.661	0.690	0.212	0.227
<i>Panel B–Above Median Tangibility Sample</i>						
Post*GOB	0.030*** (0.012)	0.031*** (0.010)	0.028*** (0.010)	0.029*** (0.010)	0.031*** (0.009)	0.035*** (0.009)
Observations	5,555	5,555	5,555	5,555	5,555	5,555
Adjusted R ²	0.682	0.739	0.664	0.714	0.139	0.225
<i>Panel C–Year of Act 1999</i>						
Post*GOB	−0.004 (0.008)	−0.001 (0.007)	−0.001 (0.008)	0.003 (0.007)	−0.003 (0.009)	0.003 (0.009)
Observations	5,861	5,861	5,861	5,861	5,861	5,861
Adjusted R ²	0.779	0.816	0.747	0.786	0.160	0.191
Controls	No	Yes	No	Yes	No	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Industry Year FE	Yes	Yes	Yes	Yes	Yes	Yes

Note:

*p<0.1; **p<0.05; ***p<0.01

This table reports results for regression: $Y_{ijt} = \alpha_i + \delta_{jt} + \theta Post_t * GOB_i + \omega X_{ijt} + u_{ijt}$. Y_{ijt} denotes the dependent variable of interest for firm i , in industry j at time t . α_i and δ_{jt} are firm and industry-year fixed effects, respectively; $Post_t$ indicates whether firm year belongs to before act (i.e., years 1999 to 2001) or after act (i.e., years 2002 to 2008) period; GOB_i indicates whether the firm has exclusive relationship with GOB in financial year 2001 or not; X_{ijt} are control variables. The coefficient of interest is θ . The standard errors are clustered at firm level and reported in the brackets. The dependent variable is *Debt/Total Assets*, *Secured Debt/Total Assets*, and *Investmetnts*. Column 1, 3, and 5 do not include controls while Column 2, 4, and 6 include controls. Panel A shows the results for the sample of firms with the below median level of tangible assets in 2001, while Panel B shows that for firms with above median level of tangible assets. Panel C gives results for assigning year of the reform as 1999. In this specification, $Post_t$ dummy takes value zero for the year 1995 to 1998, and it takes value one for the year 1999 to 2001.

Table 10: Robustness Tests–Access to Bond Markets & Others

	<i>Dependent variable: Secured Debt/Total Assets</i>				
	(1)	(2)	(3)	(4)	(5)
Post*GOB	0.024*** (0.007)	0.020*** (0.007)	−0.035 (0.037)	0.024*** (0.007)	0.022** (0.009)
Observations	10,640	11,623	672	10,248	8,662
Adjusted R ²	0.728	0.730	0.759	0.725	0.719
Controls	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes
Industry Year FE	Yes	Yes	Yes	Yes	Yes

Note:

*p<0.1; **p<0.05; ***p<0.01

This table reports results for regression: $Y_{ijt} = \alpha_i + \delta_{jt} + \theta Post_t * GOB_i + \omega X_{ijt} + u_{ijt}$. Here, Y_{ijt} denotes the dependent variable of interest for firm i , in industry j at time t . α_i and δ_{jt} are firm and industry-year fixed effects, respectively; $Post_t$ indicates whether firm year belongs to before act (i.e., years 1997 to 2001) or after act (i.e., years 2002 to 2008) period; GOB_i indicates whether the firm has exclusive relationship with GOB in financial year 2001 or not; X_{ijt} are control variables. The coefficient of interest is θ . The standard errors are clustered at firm level and reported in the brackets. The dependent variable is *Secured Debt/Total Assets*. In Column 1, firms with access to bond market are removed, while in Column 2, Zombie firms are added to the baseline sample. Column 3 gives results for only Zombie firms. In Column 4, the sample is restricted to firms that are present in the last year of sample. In Column 5, distance-to-default is added as control in place of *Z-Score*.

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Appendices

A. Appendix

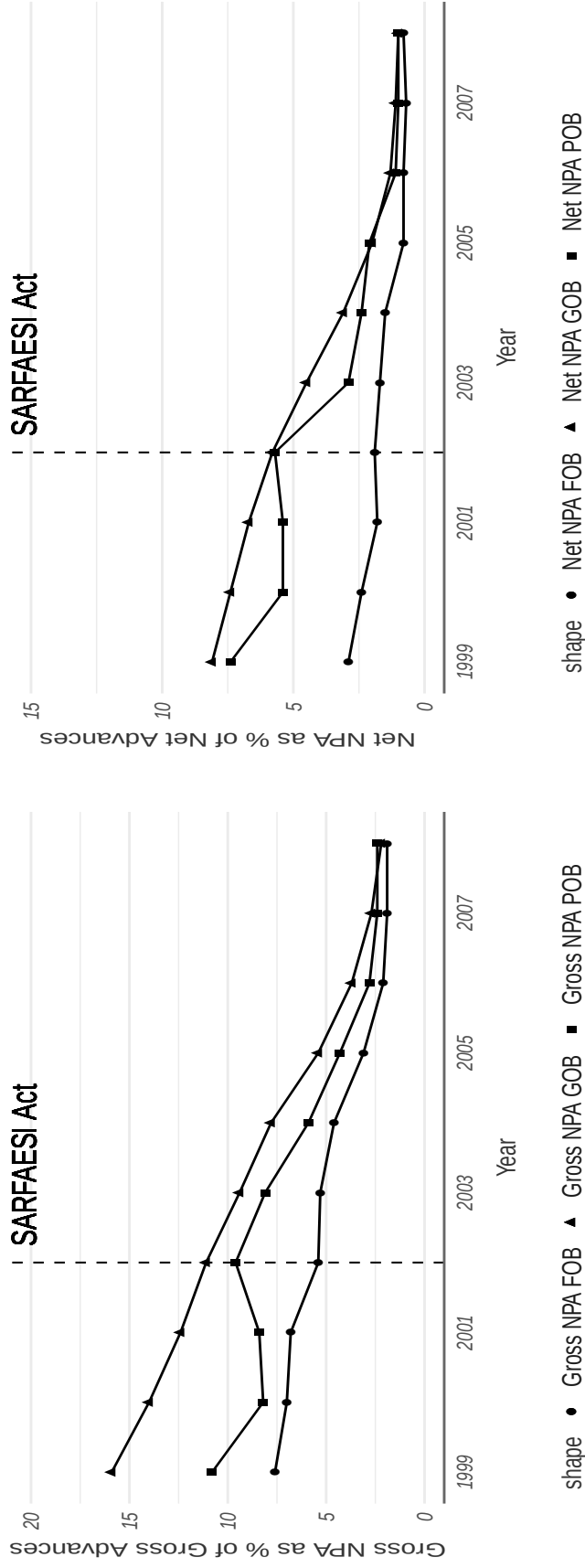


Fig. A1. NPAs Bank Group Wise

This figure shows the plots of gross NPAs as a percentage of gross advances (left panel) and net NPAs as a percentage of net advances (right panel) for the period 1999 to 2008 for all the scheduled commercial banks operating in India by their ownership group. The three lines represent three groups: government owned banks (GOB), privately owned banks (POB), foreign owned banks (FOB). The data is taken from the Reserve Bank of India's publicly available web database.

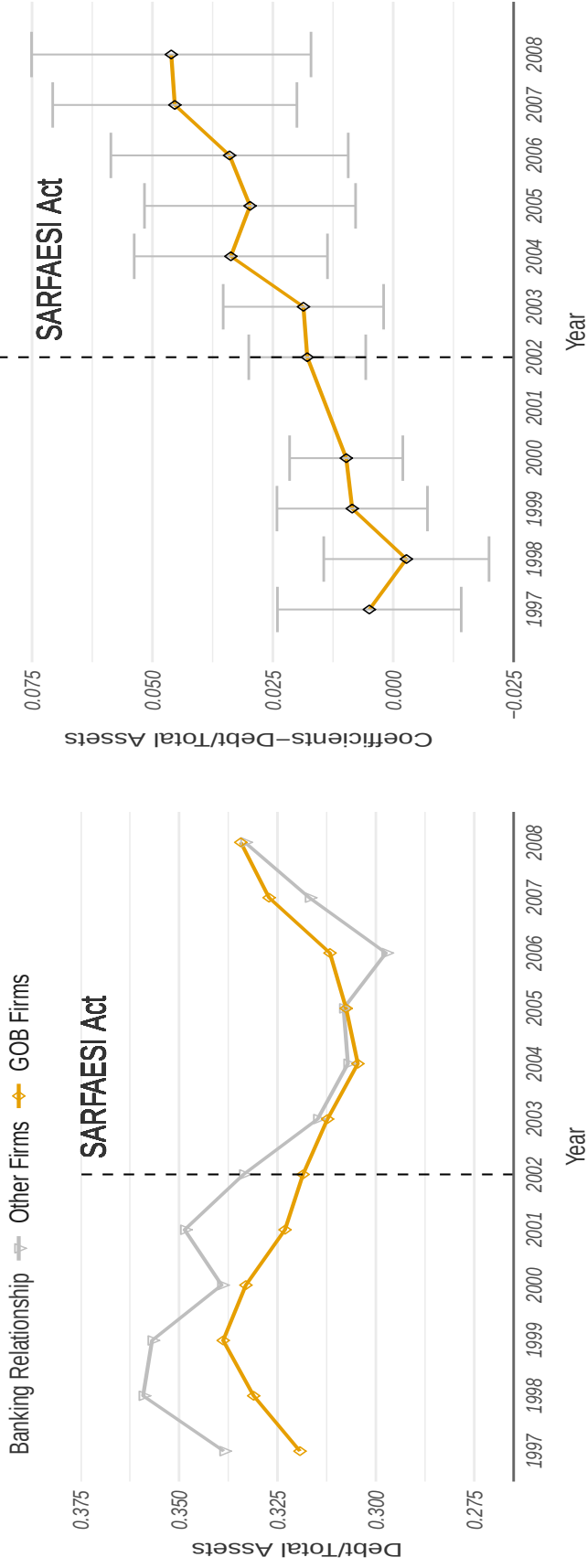


Fig. A2. Debt as a proportion of Total Assets

This figure is for *Debt/Total Assets*. Left panel shows the plot of group wise yearly average for GOB firms and Other firms. GOB firms are firms having exclusive relationship with GOB in the year before the reform. The rest of the firms belong to Other group. Right panel shows the plot of coefficients θ_{Year} for the non-parametric difference-in-difference regression: $Y_{ijt} = \alpha_i + \delta_{jt} + \sum_{Year=1997}^{2008} \theta_{Year} Year * GOB_i + u_{ijt}$. Here, α_i and δ_{jt} are firm and industry-year fixed effects respectively; $Post_t$ indicates whether firm year belongs to before act (1997 to 2001) or after act (2002 to 2008) period; GOB_i indicates whether the firm has exclusive relationship with GOBs in financial year 2001 or not. The standard errors are clustered at firm level. Error bars show 95% confidence interval.

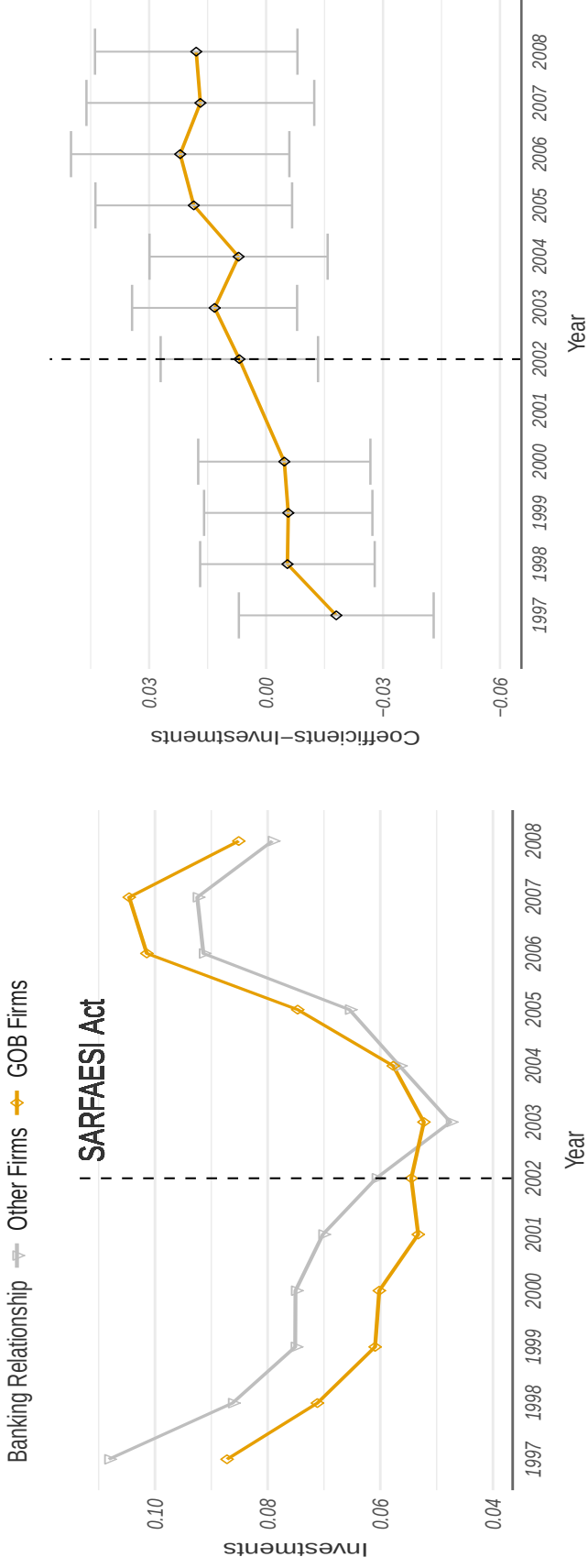
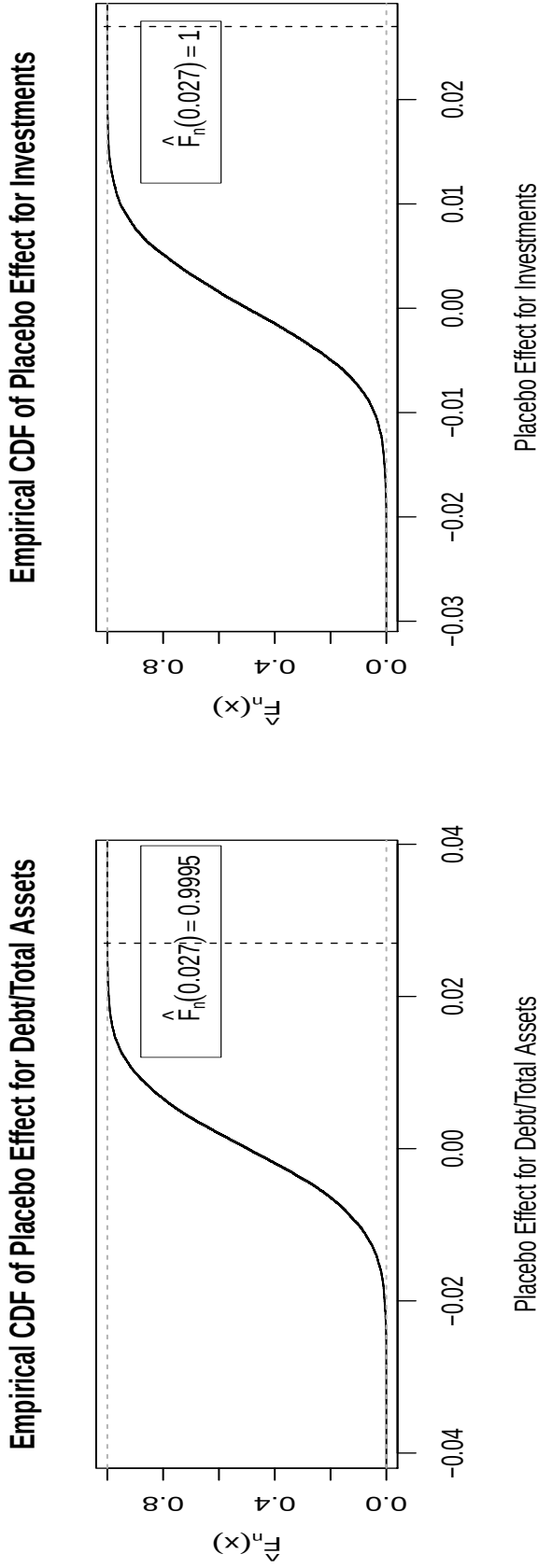


Fig. A3. Investments as a proportion of Total Assets In Previous Year

This figure shows the plots for *Investments*. The left panel gives group wise yearly average of investments for GOB firms and Other firms. The right panel gives the plot of coefficients θ_{Year} for the non-parametric difference-in-difference regression: $Y_{ijt} = \alpha_i + \delta_{jt} + \sum_{Year=1997}^{2008} \theta_{Year} * GOB_i + u_{ijt}$. α_i and δ_{jt} are firm and industry-year fixed effects respectively; $Post_t$ indicates whether firm year belongs to before act (i.e., years 1997 to 2001) or after act (i.e., years 2002 to 2008) period; GOB_i indicates whether the firm has exclusive relationship with GOBs in financial year 2001 or not. The standard errors are clustered at firm level. Error bars show 95% confidence interval. Sample Period is 1999 to 2008.

Fig. A4. Empirical CDF of Placebo Effect for Debt/Total Assets & Investments



This figure shows the empirical CDF of placebo effects calculated for the ratio of debt to total assets (left panel) and investments (right panel). The CDF is constructed from 10000 estimates of differential effect i.e., interaction term using following equation for the baseline sample* $Y_{ijt} = \alpha_i + \delta_{jt} + \theta Post_t * GOB_i + \omega X_{ijt} + u_{ijt}$. No parametric smoothing is used: the empirical CDF appears smooth due to the 10000 data points used to construct it. The vertical line in left and right panel denote the estimated effect presented in appendix Table A4 Column 4 and Table 7 Column 4 respectively.

Table A1: Variable Description

Variable Name	Description
Total Assets	Book value of total assets.
Debt	It includes all borrowings plus preference share outstanding.
Log Total Assets	Natural logarithm of book value of total assets.
Interest Cov Ratio	PBITDA/Interest Expenses
Interest Costs Ratio	Interest Expenses/Average Borrowings
Tangibility	(Net book value of property, plant and equipment + Net book value of other fixed assets) / Total assets.
Debt/Total Assets	Ratio of Debt to Total Assets. Both variables are taken from the prowess.
Secured Debt/Total Assets	Ratio of Secured Borrowings to Total Assets.
Investments	Change in gross fixed assets, scaled by total assets in the previous year.
Cashflow	Net cash flow from operating activities scaled by total assets in the previous year.
Tobin-Q	(Total Assets-Networth+ Market Value of Equity)/Total Assets
GOB	A dummy variable that takes a value of 1 if firm borrows exclusively from government owned banks in financial year 2001 and 0 otherwise.
Post	A dummy variable that takes a value of 1 for financial year 2002 and later, and 0 otherwise.
Zombie Firm	I classify a firm as Zombie if all of the following three criteria are satisfied: 1. Interest coverage ratio (ICR) less than equal to 1. 2. Leverage (total debt to total assets) of greater than 0.20. 3. Taken additional loan in year t.

Table A2: Sample Industry Details

FF Code	FF Industry Name	Obs	Firms	GOB
1	Food	796	114	63
2	Alcohol	45	7	5
3	Smoke	4	1	0
4	Games	168	24	6
5	Books	20	4	2
6	Household	329	45	18
7	Apparel	151	19	13
8	Healthcare	806	106	60
9	Chemicals	1658	209	136
10	Textile	1072	136	76
11	Construction	904	114	62
12	Steel	680	87	57
13	Fabricated Products and Machinery	675	99	49
14	Electrical Equipment	381	51	27
15	Automobiles and Trucks	401	56	29
16	Transport equipment	20	3	2
17	Mines	81	14	10
19	Oil	83	13	1
20	Utilities	41	5	2
21	Telcm	30	4	2
22	Services	563	104	40
23	Business Equipment	169	20	12
24	Paper	194	29	23
25	Transport	131	18	7
26	Wholesale	1039	156	96
27	Retail	90	11	4
28	Meals	137	19	8
34	Diversified	274	36	16
	Miscellaneous	9	1	0

This table reports Fama-French industry details for the full sample. Prowess industries are mapped to Fama-French industry codes using data from Srinivasan and Thamby (2017). Column 1 and 2 give Fama-French industry code, and name respectively. Column 3 gives total number of observations for each industry. Column 4 shows the number of firms in each industry and Column 5 shows how many firms among them are GOB firms. GOB firms are firms that have an exclusive relationship with GOBs in the financial year 2001.

Table A3: Sample Industry Year Details

FF_Code	FF_Name	1997	1998	1999	2000	2001	2002	2003	2004	2005	2006	2007	2008
1	Food	82	89	85	86	73	62	58	55	54	55	53	44
2	Alcohol	7	6	6	6	4	5	2	2	2	1	2	2
3	Smoke	1	1	1	1	0	0	0	0	0	0	0	0
4	Games	14	11	13	11	17	16	14	15	14	13	15	15
5	Books	3	3	2	3	3	1	0	1	1	1	1	1
6	Household	37	39	38	32	31	25	20	21	20	21	22	23
7	Apparel	18	14	12	13	13	10	12	11	11	13	12	12
8	Healthcare	72	73	74	79	75	69	63	58	62	62	60	59
9	Chemicals	171	175	176	176	140	133	124	115	111	114	114	109
10	Textile	115	114	112	122	95	85	75	63	68	76	76	71
11	Construction	89	87	87	89	79	69	68	64	70	68	70	64
12	Steel	70	68	65	69	58	52	48	50	52	49	50	49
13	Fabricated Products and Machinery	86	86	72	77	60	44	44	39	39	48	44	36
14	Electrical Equipment	44	44	42	45	36	30	23	22	23	25	25	22
15	Automobiles and Trucks	53	52	49	45	29	24	23	20	23	27	33	23
16	Transport equipment	3	3	2	3	3	2	3	0	0	1	0	0
17	Mines	8	9	10	10	7	6	6	7	5	5	5	3
19	Oil	7	8	8	10	6	5	6	2	4	10	8	9
20	Utilities	5	5	3	5	4	3	2	3	3	3	3	2
21	Telcm	3	3	2	4	4	3	3	2	2	2	2	0
22	Services	44	47	49	52	59	55	49	39	42	42	47	38
23	Business Equipment	14	14	15	18	16	16	15	14	13	10	13	11
24	Paper	21	23	20	26	19	12	14	13	11	12	11	12
25	Transport	11	11	10	13	11	10	11	10	10	11	12	11
26	Wholesale	108	110	97	115	99	89	79	73	68	71	65	65
27	Retail	7	7	6	9	6	8	7	8	8	7	8	9
28	Meals	16	16	16	15	14	13	9	7	8	8	8	7
34	Diversified	31	30	30	28	24	23	19	19	16	20	17	17
	Miscellaneous	1	1	0	0	1	1	1	1	1	1	1	0

This table reports yearly details of number of firms in each Fama-French industry. Prowess industries are mapped to Fama-French industry codes using data from Srinivasan and Thampy (2017).

Table A4: Results For Debt/ Total Assets

	<i>Dependent variable:</i>						
	Debt/Total Assets						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Log Sales _{t-1}		0.032*** (0.002)	0.016*** (0.004)	0.015*** (0.004)	0.018*** (0.005)	0.013*** (0.004)	0.012*** (0.004)
Tangibility _{t-1}		0.302*** (0.018)	0.061*** (0.020)	0.060*** (0.021)	0.093*** (0.022)	0.033* (0.020)	0.060*** (0.021)
Profitability _{t-1}		-0.386*** (0.037)	-0.380*** (0.031)	-0.374*** (0.032)	-0.254*** (0.040)	-0.209*** (0.029)	-0.368*** (0.032)
Z-Score						-0.029*** (0.002)	
# Banks							0.005*** (0.001)
Post*GOB	0.020** (0.009)	0.018** (0.009)	0.025*** (0.008)	0.027*** (0.008)	0.022*** (0.008)	0.024*** (0.008)	0.029*** (0.008)
Post*Log Sales _{t-1}					-0.004 (0.003)		
Post*Tangibility _{t-1}					-0.068*** (0.021)		
Post*Profitability _{t-1}					-0.228*** (0.053)		
Post*Z-Score						0.001 (0.002)	
Observations	10,951	10,951	10,951	10,951	10,951	10,951	10,951
Adjusted R ²	0.005	0.165	0.722	0.728	0.732	0.754	0.729
Controls	No	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	No	No	Yes	Yes	Yes	Yes	Yes
Industry Year FE	No	No	No	Yes	Yes	Yes	Yes

Note:

*p<0.1; **p<0.05; ***p<0.01

This table reports results for the regression: $Y_{ijt} = \alpha_i + \delta_{jt} + \theta Post_t * GOB_i + \omega X_{ijt} + u_{ijt}$. Here, Y_{ijt} denotes *Debt/Total Assets* for firm i , in industry j at time t . α_i and δ_{jt} are firm and industry-year fixed effects respectively; $Post_t$ indicates whether firm year belongs to before act (1997 to 2001) or after act (2002 to 2008) period; GOB_i indicates whether the firm has exclusive relationship with GOB in financial year 2001 or not; X_{ijt} are control variables. The standard errors are clustered at firm level and reported in the brackets. Column 1 reports results without controls and fixed effects. Controls are added in Column 2, while firm and industry year fixed effects are added in Column 3 and 4, respectively. In Column 5, I add interaction of *Post* with Rajan and Zingales (1995) controls. I control for *Z-Score* and *# Banks* in Column 6 and 7, respectively.

Table A5: Bias Corrected Matching Estimator Including Government Owned Firms

	Debt/ Total Assets	Secured Debt/ Total Assets	Investments
<i>Panel A: Full Matching</i>			
GOB Firms* After - Before	0.003	-0.003	0.001
Other Firms* After - Before	-0.024	-0.030	-0.030
Bias Corrected Matching Estimate	0.032***	0.026***	0.030***
Standard Error	0.011	0.010	0.008
No. of Firms	870	870	870
<i>Panel B: Restricted Matching-# Banks $\leq$ Max(GOB Firms' #Banks)</i>			
GOB Firms* After - Before	0.003	-0.003	0.001
Other Firms* After - Before	-0.022	-0.028	-0.030
Bias Corrected Matching Estimate	0.030***	0.024**	0.029***
Standard Error	0.011	0.010	0.008
No. of Firms	852	852	852
<i>Note:</i>		*p<0.1; **p<0.05; ***p<0.01	

This table presents the matching estimates of the change in *Debt/Total Assets*, *Secured Debt/Total Assets* and *Investments* around SARFAESI Act. The first row in both the panels contains the differences in the outcome variable for GOB firms between the pre-reform (1997-2001) and the post-reform (2002-2008) period, and the second row contains the differences in the outcome variables for matched control group firms over the same period. The third row contains the estimate for the average treatment effect on the treated (ATT) based on the bias-corrected matching estimator using Mahalanobis metric (Abadie and Imbens (2002)). Each GOB firm is matched to four firms from Other group (1-to-4 Match). Panel A presents the results for all firms in the sample. In Panel B matching for the control group is restricted to the firms that have *# Banks* less than or equal to maximum *# Banks* among GOB firms.

Table A6: Robustness Using Control Variables in Levels

	<i>Dependent variable:</i>						
	Secured Debt/Total Assets						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Log Sales		0.027*** (0.002)	0.014*** (0.004)	0.015*** (0.005)	0.025*** (0.005)	0.013*** (0.004)	0.015*** (0.005)
Tangibility		0.282*** (0.017)	0.152*** (0.023)	0.152*** (0.023)	0.187*** (0.024)	0.084*** (0.024)	0.152*** (0.023)
Profitability		-0.327*** (0.035)	-0.319*** (0.028)	-0.314*** (0.029)	-0.212*** (0.037)	-0.017 (0.039)	-0.312*** (0.029)
Z-Score						-0.028*** (0.003)	
# Banks							0.001 (0.001)
Post*GOB	0.021** (0.008)	0.020** (0.008)	0.026*** (0.008)	0.027*** (0.008)	0.017** (0.008)	0.024*** (0.007)	0.028*** (0.008)
Post*Log Sales					-0.009*** (0.003)		
Post*Tangibility					-0.064*** (0.020)		
Post*Profitability					-0.214*** (0.052)		
Post*Z-Score						0.003 (0.002)	
Observations	10,304	10,304	10,304	10,304	10,304	10,304	10,304
Adjusted R ²	0.005	0.166	0.703	0.707	0.712	0.726	0.707
Controls	No	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	No	No	Yes	Yes	Yes	Yes	Yes
Industry Year FE	No	No	No	Yes	Yes	Yes	Yes

Note:

*p<0.1; **p<0.05; ***p<0.01

This table reports results for the regression: $Y_{ijt} = \alpha_i + \delta_{jt} + \theta Post_t * GOB_i + \omega X_{ijt} + u_{ijt}$. Here, Y_{ijt} denotes *Secured Debt/Total Assets* for firm i , in industry j at time t . α_i and δ_{jt} are firm and industry-year fixed effects respectively; $Post_t$ indicates whether firm year belongs to before act (1997 to 2001) or after act (2002 to 2008) period; GOB_i indicates whether the firm has exclusive relationship with GOB in financial year 2001 or not; X_{ijt} are control variables. The standard errors are clustered at firm level and reported in the brackets. Column 1 reports results without controls and fixed effects. Rajan and Zingales (1995) controls are added in Column 2, while firm and industry year fixed effects are added in Column 3 and 4, respectively. In Column 5, I add interaction of $Post$ with Rajan and Zingales (1995) controls. I control for Z -Score and $\#$ Banks in Column 6 and 7, respectively.

Table A7: Robustness Tests–1

	<i>Dependent variable:</i>					
	Debt/Total Assets		Secured Debt/Total Assets		Investments	
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A–Restricting Sample Till 2004</i>						
Post*GOB	0.017** (0.008)	0.016** (0.008)	0.015* (0.008)	0.014** (0.007)	0.015** (0.007)	0.022*** (0.007)
Observations	7,943	7,943	7,943	7,943	7,943	7,943
Adjusted R ²	0.795	0.822	0.768	0.793	0.148	0.189
<i>Panel B–Year of Act–2003</i>						
Post*GOB	0.028*** (0.009)	0.026*** (0.008)	0.028*** (0.008)	0.026*** (0.007)	0.020*** (0.006)	0.026*** (0.007)
Observations	10,951	10,951	10,951	10,951	10,951	10,951
Adjusted R ²	0.715	0.755	0.693	0.728	0.154	0.203
<i>Panel C–Remove Firms Switching GOB Category After Act</i>						
Post*GOB	0.027*** (0.010)	0.024*** (0.009)	0.028*** (0.009)	0.023*** (0.008)	0.017** (0.007)	0.030*** (0.007)
Observations	7,662	7,662	7,662	7,662	7,662	7,662
Adjusted R ²	0.724	0.765	0.716	0.750	0.138	0.182
Controls	No	Yes	No	Yes	No	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Industry Year FE	Yes	Yes	Yes	Yes	Yes	Yes

Note:

*p<0.1; **p<0.05; ***p<0.01

This table reports results for regression: $Y_{ijt} = \alpha_i + \delta_{jt} + \theta Post_t * GOB_i + \omega X_{ijt} + u_{ijt}$. Here, Y_{ijt} denotes the dependent variable of interest for firm i , in industry j at time t . α_i and δ_{jt} are firm and industry-year fixed effects, respectively; $Post_t$ indicates whether firm year belongs to before act (1997 to 2001) or after act (2002 to 2008) period; GOB_i indicates whether the firm has exclusive relationship with GOB in financial year 2001 or not; X_{ijt} are control variables. The coefficient of interest is θ . The standard errors are clustered at firm level and reported in the brackets. The dependent variables are *Debt/Total Assets*, *Secured Debt/Total Assets*, and *Investmetnts*. Column 1, 3, and 5 does not include controls while Column 2, 4, and 6 does. Panel A gives results for restricting the sample to 2004 while Panel B shows the results by taking year of act as 2003. Panel C gives results after excluding firms switching GOB categorization after the reform.

Table A8: Robustness Tests–2-Including *TobinQ* in Controls

	<i>Dependent variable:</i>		
	Sec.Plus Unsec. Debt/	Secured Debt/	Investments
	Total Assets	Total Assets	
	(1)	(2)	(3)
Log Sales _{t-1}	0.011** (0.005)	0.009** (0.004)	
Tangibility _{t-1}	0.005 (0.021)	0.030 (0.018)	
Profitability _{t-1}	-0.251*** (0.035)	-0.139*** (0.027)	
Cashflow			0.136*** (0.023)
Log Total Assets			0.070*** (0.006)
Debt/Total Assets ₋₁			-0.130*** (0.020)
TobinQ _{t-1}	0.00002 (0.004)	-0.0003 (0.002)	0.008** (0.003)
Z-Score	-0.029*** (0.002)	-0.025*** (0.002)	
# Banks	0.005*** (0.001)	0.002 (0.001)	
Post*GOB	0.021** (0.009)	0.024*** (0.008)	0.027*** (0.007)
Observations	8,245	10,189	10,189
Adjusted R ²	0.751	0.734	0.199
Controls	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes
Industry Year FE	Yes	Yes	Yes

Note:

*p<0.1; **p<0.05; ***p<0.01

This table reports results for regression: $Y_{ijt} = \alpha_i + \delta_{jt} + \theta Post_t * GOB_i + \omega X_{ijt} + u_{ijt}$. Here, Y_{ijt} denotes the dependent variable of interest for firm i , in industry j at time t . α_i and δ_{jt} are firm and industry-year fixed effects respectively; $Post_t$ indicates whether firm year belongs to before act (1997 to 2001) or after act (2002 to 2008) period; GOB_i indicates whether the firm has exclusive relationship with GOB in financial year 2001 or not; X_{ijt} are control variables which includes *TobinQ*. The coefficient of interest is θ . The standard errors are clustered at firm level and reported in the brackets. The dependent variables are *Sec. Plus Unsec. Debt/Total Assets*, *Secured Debt/Total Assets*, and *Investmetnts*.

Table A9: Robustness Tests–3–Including Zombie Firms

	<i>Dependent variable:</i>					
	Debt/Total Assets		Secured Debt/Total Assets		Investments	
	(1)	(2)	(3)	(4)	(5)	(6)
Post*GOB	0.023*** (0.008)	0.021*** (0.008)	0.022*** (0.007)	0.020*** (0.007)	0.019*** (0.006)	0.025*** (0.006)
Observations	11,623	11,623	11,623	11,623	11,623	11,623
Adjusted R ²	0.713	0.756	0.692	0.730	0.152	0.200
Controls	No	Yes	No	Yes	No	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Industry Year FE	Yes	Yes	Yes	Yes	Yes	Yes

Note:

*p<0.1; **p<0.05; ***p<0.01

This table reports results for regression: $Y_{ijt} = \alpha_i + \delta_{jt} + \theta Post_t * GOB_i + \omega X_{ijt} + u_{ijt}$ after adding Zombie firms to the baseline sample. Here, Y_{ijt} denotes the dependent variable of interest for firm i , in industry j at time t . α_i and δ_{jt} are firm and industry-year fixed effects, respectively; $Post_t$ indicates whether firm year belongs to before act (1997 to 2001) or after act (2002 to 2008) period; GOB_i indicates whether the firm has exclusive relationship with GOB in financial year 2001 or not; X_{ijt} are control variables. The coefficient of interest is θ . The standard errors are clustered at firm level and reported in the brackets. The dependent variables are *Debt/Total Assets*, *Secured Debt/Total Assets*, and *Investmetnts*. Column 1, 3, and 5 does not include controls while Column 2, 4, and 6 does.

Table A10: Robustness Test–4–Interest Rate Based Definition of Zombie Firms

	<i>Dependent variable:</i>					
	Debt/Total Assets		Secured Debt/Total Assets		Investments	
	(1)	(2)	(3)	(4)	(5)	(6)
Post*GOB	0.023** (0.009)	0.021** (0.009)	0.022** (0.009)	0.019** (0.008)	0.019*** (0.007)	0.024*** (0.007)
Observations	9,201	9,201	9,201	9,201	9,201	9,201
Adjusted R ²	0.716	0.756	0.696	0.733	0.158	0.203
Controls	No	Yes	No	Yes	No	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Industry Year FE	Yes	Yes	Yes	Yes	Yes	Yes

Note:

*p<0.1; **p<0.05; ***p<0.01

This table reports results for regression: $Y_{ijt} = \alpha_i + \delta_{jt} + \theta Post_t * GOB_i + \omega X_{ijt} + u_{ijt}$ after removing Zombie firms with interest rate based definition. A firm is defined as a Zombie if its $Interest\ Cost = \frac{Interest\ Expense}{Avg.\ Borrowings}$ is less than prime lending rate of State Bank of India in 2001. Here, Y_{ijt} denotes the dependent variable of interest for firm i , in industry j at time t . α_i and δ_{jt} are firm and industry-year fixed effects, respectively; $Post_t$ indicates whether firm year belongs to before act (1997 to 2001) or after act (2002 to 2008) period; GOB_i indicates whether the firm has exclusive relationship with GOB in financial year 2001 or not; X_{ijt} are control variables. The coefficient of interest is θ . The standard errors are clustered at firm level and reported in the brackets. The dependent variables are *Debt/Total Assets*, *Secured Debt/Total Assets*, and *Investmetnts*. Column 1, 3, and 5 does not include controls while Column 2, 4, and 6 does.

Table A11: Comparison of Sample with Vig (2013)

Details	Missing	Obs Left
Obs. excl. financial firms, firms years with negative net worth or sales	–	53,306
Non-availability of debt and controls variables used in Vig (2013)**	25,052	28,254
Non-availability of information on firm's bankers	16,142	12,112***
Non-availability on some variables not used in Vig (2013)	246	11,866
Removing non-financial Govt. Owned firms	243	11,623
Removing Zombie firms	672	10,951
Final Sample		10,951

**Except TobinQ

***Total of 12,112 observations for firms' banking relationship are comparable to Vig's (2013) 9,456 observations for number of banks in his Table III-A.